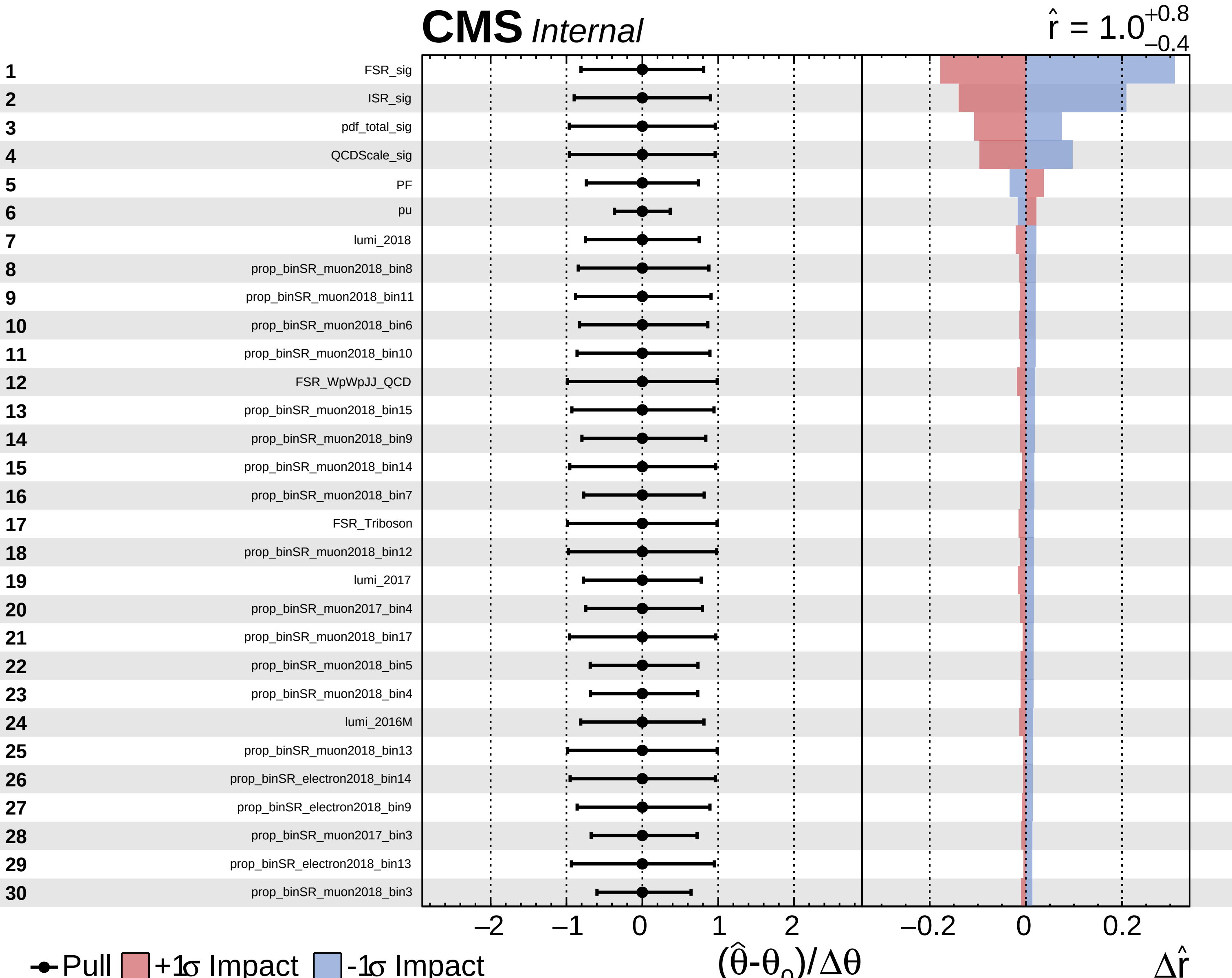
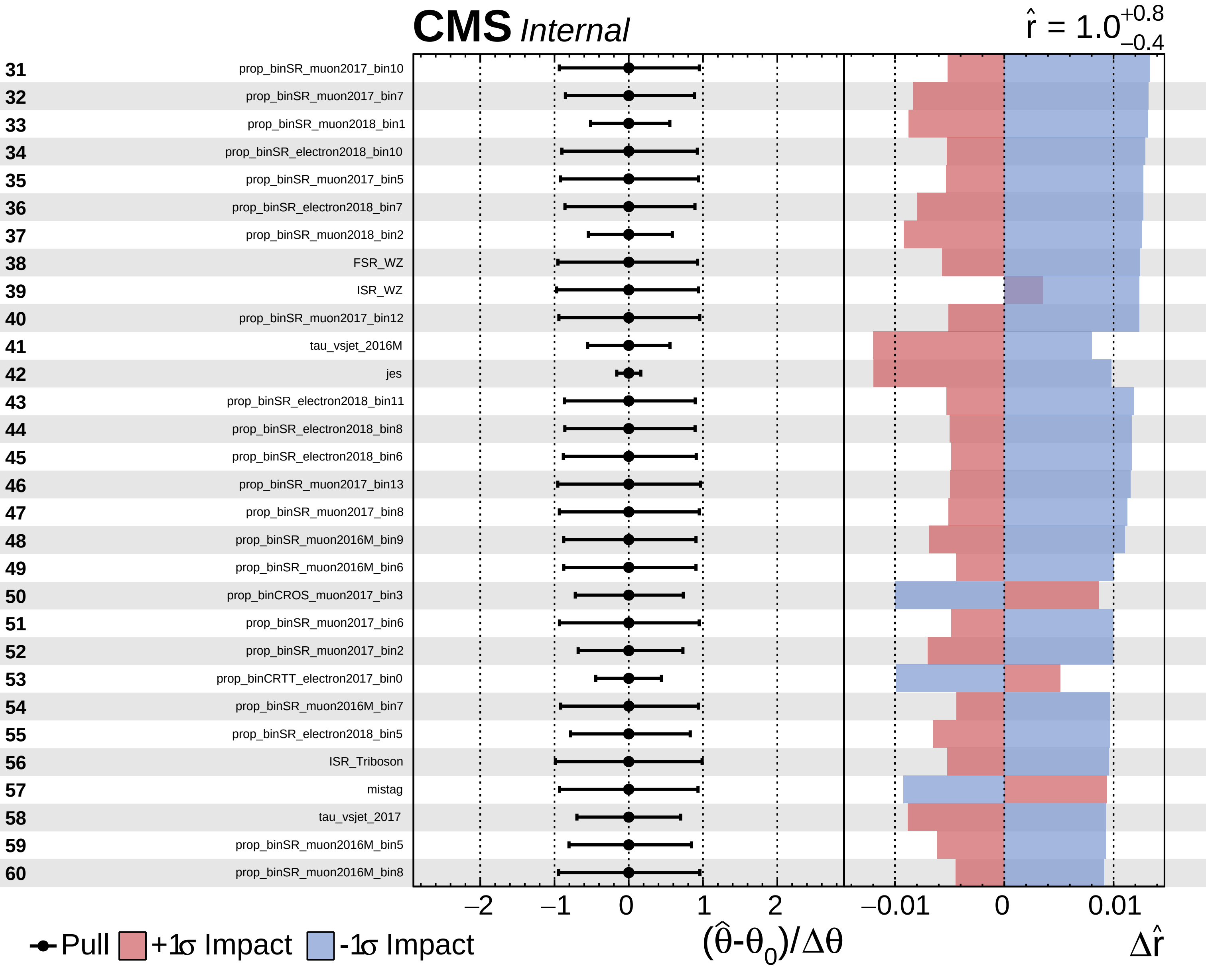
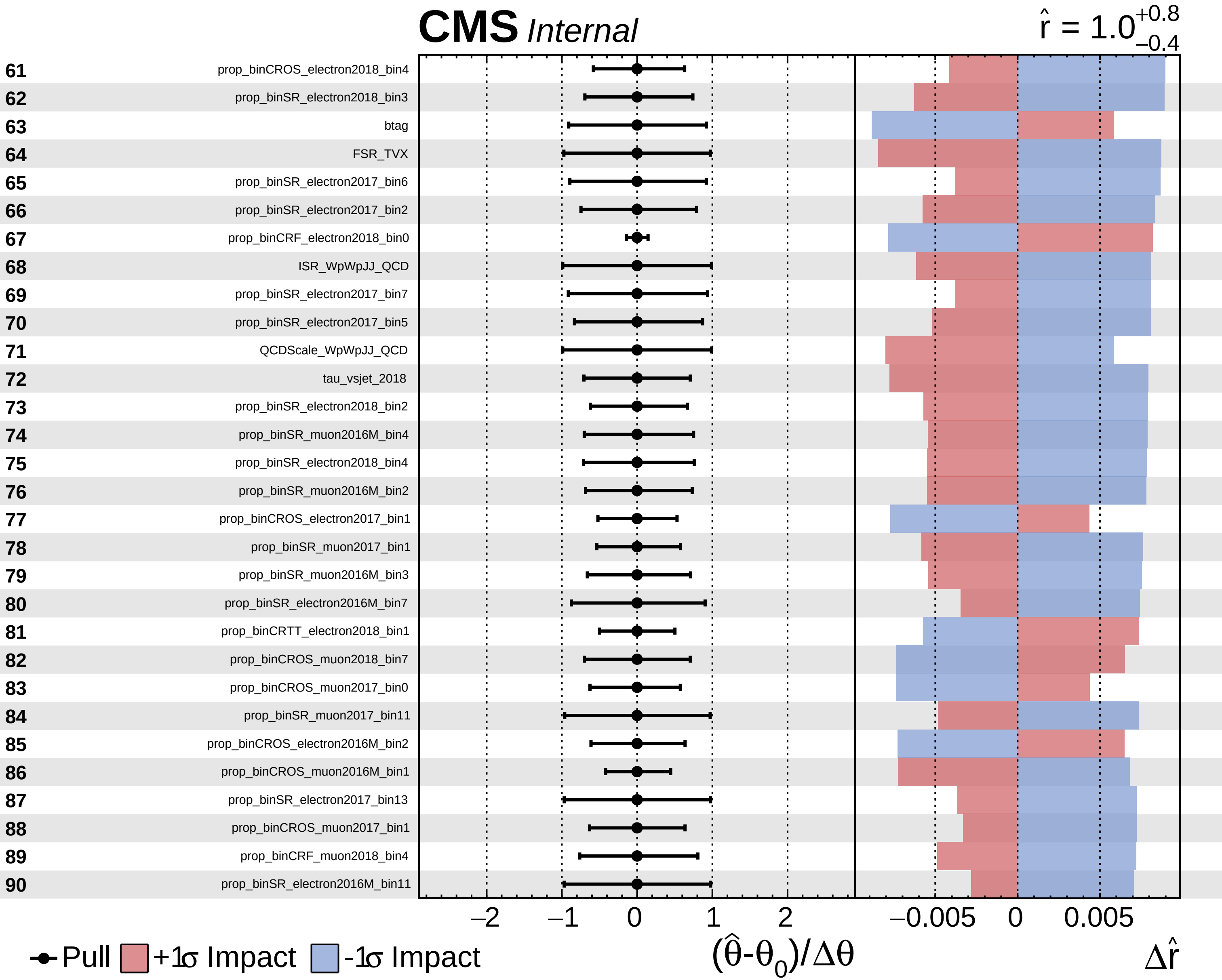
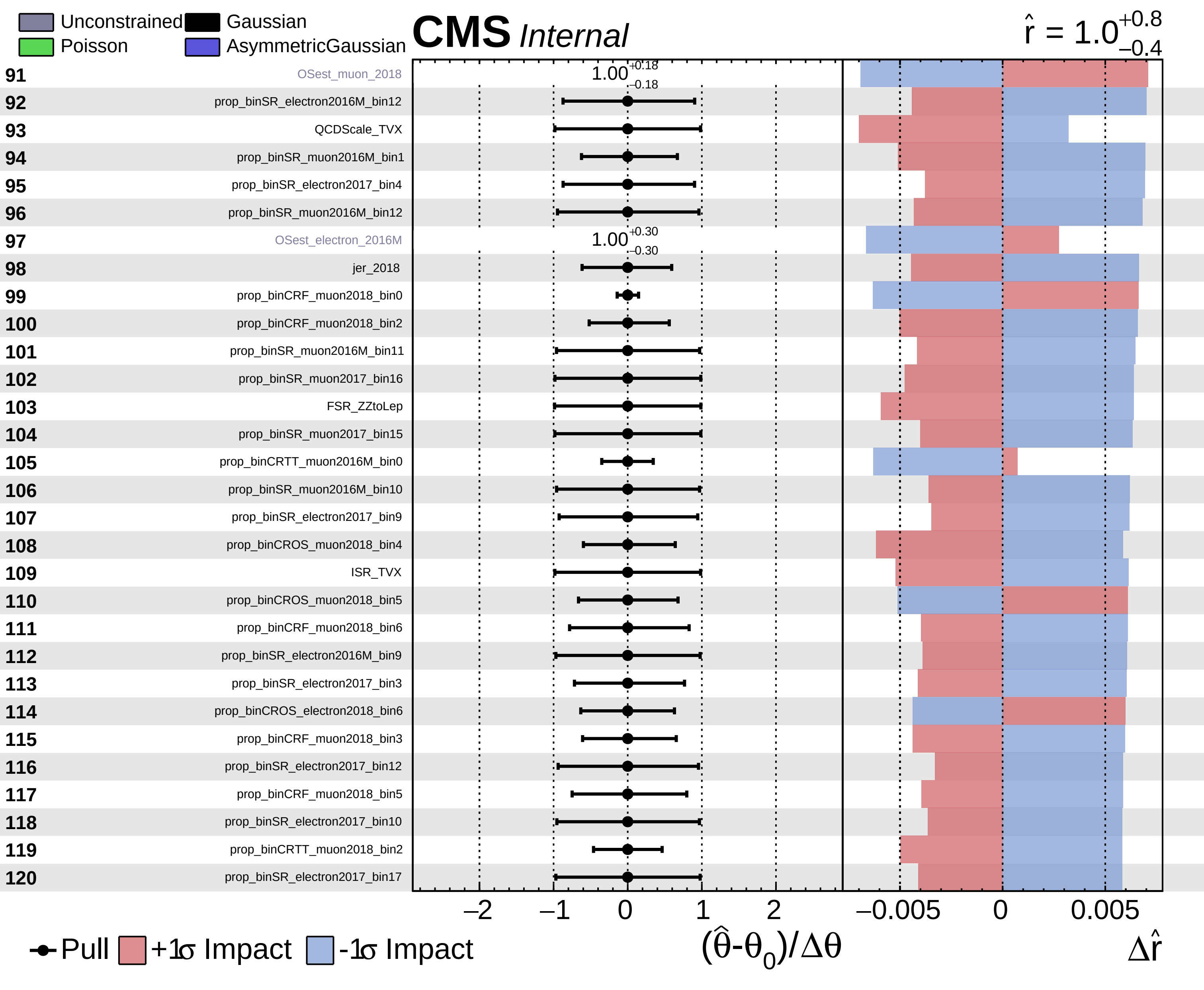

CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
1 FSR_sig
2 ISR_sig
3 pdf_total_sig
4 QCDScale_sig
5 PF
6 pu
7 lumi_2018
8 prop_binSR_muon2018_bin8
9 prop_binSR_muon2018_bin11
10 prop_binSR_muon2018_bin6
11 prop_binSR_muon2018_bin10
12 FSR_WpWpJJ_QCD
13 prop_binSR_muon2018_bin15
14 prop_binSR_muon2018_bin9
15 prop_binSR_muon2018_bin14
16 prop_binSR_muon2018_bin7
17 FSR_Triboson
18 prop_binSR_muon2018_bin12
19 lumi_2017
20 prop_binSR_muon2017_bin4
21 prop_binSR_muon2018_bin17
22 prop_binSR_muon2018_bin5
23 prop_binSR_muon2018_bin4
24 lumi_2016M
25 prop_binSR_muon2018_bin13
26 prop_binSR_electron2018_bin14
27 prop_binSR_electron2018_bin9
28 prop_binSR_muon2017_bin3
29 prop_binSR_electron2018_bin13
30 prop_binSR_muon2018_bin3
−2 −1 0 1 2
−0.2 0 0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

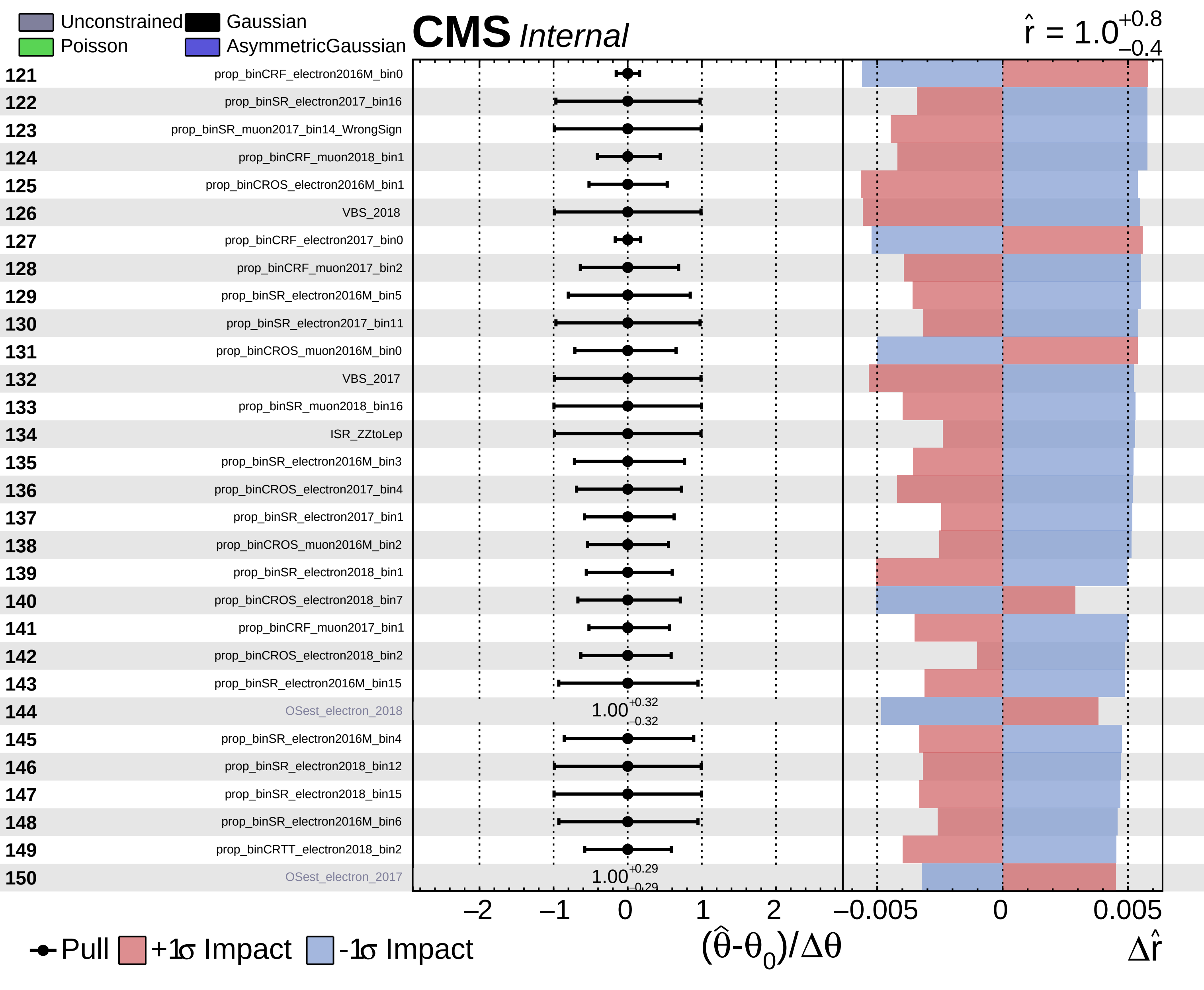

CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
31 prop_binSR_muon2017_bin10
32 prop_binSR_muon2017_bin7
33 prop_binSR_muon2018_bin1
34 prop_binSR_electron2018_bin10
35 prop_binSR_muon2017_bin5
36 prop_binSR_electron2018_bin7
37 prop_binSR_muon2018_bin2
38 FSR_WZ
39 ISR_WZ
40 prop_binSR_muon2017_bin12
41 tau_vsjet_2016M
42 jes
43 prop_binSR_electron2018_bin11
44 prop_binSR_electron2018_bin8
45 prop_binSR_electron2018_bin6
46 prop_binSR_muon2017_bin13
47 prop_binSR_muon2017_bin8
48 prop_binSR_muon2016M_bin9
49 prop_binSR_muon2016M_bin6
50 prop_binCROS_muon2017_bin3
51 prop_binSR_muon2017_bin6
52 prop_binSR_muon2017_bin2
53 prop_binCRTT_electron2017_bin0
54 prop_binSR_muon2016M_bin7
55 prop_binSR_electron2018_bin5
56 ISR_Triboson
57 mistag
58 tau_vsjet_2017
59 prop_binSR_muon2016M_bin5
60 prop_binSR_muon2016M_bin8
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



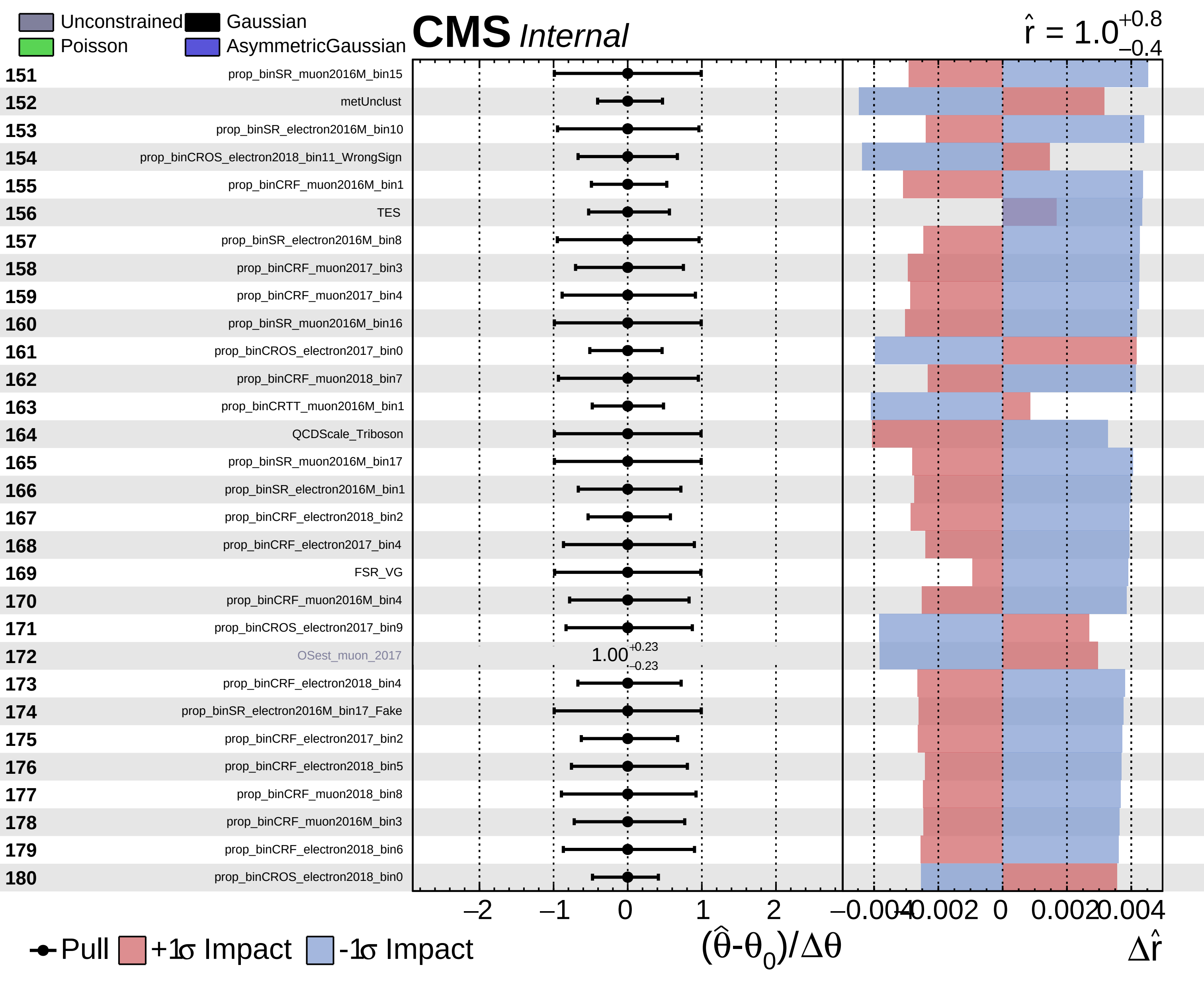
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
61 prop_binCROS_electron2018_bin4
62 prop_binSR_electron2018_bin3
63 btag
64 FSR_TVX
65 prop_binSR_electron2017_bin6
66 prop_binSR_electron2017_bin2
67 prop_binCRF_electron2018_bin0
68 ISR_WpWpJJ_QCD
69 prop_binSR_electron2017_bin7
70 prop_binSR_electron2017_bin5
71 QCDScale_WpWpJJ_QCD
72 tau_vsjet_2018
73 prop_binSR_electron2018_bin2
74 prop_binSR_muon2016M_bin4
75 prop_binSR_electron2018_bin4
76 prop_binSR_muon2016M_bin2
77 prop_binCROS_electron2017_bin1
78 prop_binSR_muon2017_bin1
79 prop_binSR_muon2016M_bin3
80 prop_binSR_electron2016M_bin7
81 prop_binCRTT_electron2018_bin1
82 prop_binCROS_muon2018_bin7
83 prop_binCROS_muon2017_bin0
84 prop_binSR_muon2017_bin11
85 prop_binCROS_electron2016M_bin2
86 prop_binCROS_muon2016M_bin1
87 prop_binSR_electron2017_bin13
88 prop_binCROS_muon2017_bin1
89 prop_binCRF_muon2018_bin4
90 prop_binSR_electron2016M_bin11
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

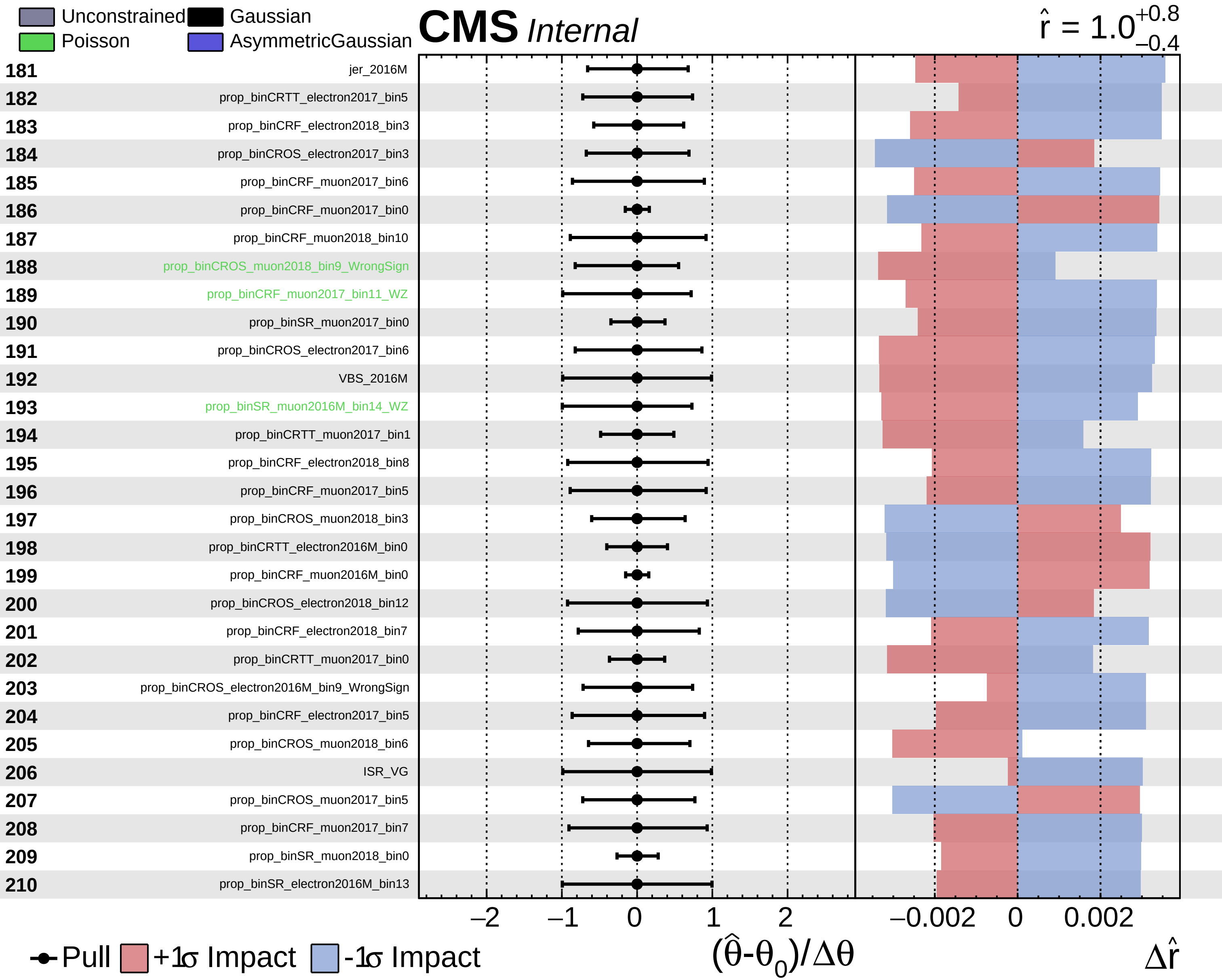


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
91 OSest_muon_2018
$1.00^{+0.18}_{-0.18}$
92 prop_binSR_electron2016M_bin12
93 QCDScale_TVX
94 prop_binSR_muon2016M_bin1
95 prop_binSR_electron2017_bin4
96 prop_binSR_muon2016M_bin12
97 OSest_electron_2016M
$1.00^{+0.30}_{-0.30}$
98 jer_2018
99 prop_binCRF_muon2018_bin0
100 prop_binCRF_muon2018_bin2
101 prop_binSR_muon2016M_bin11
102 prop_binSR_muon2017_bin16
103 FSR_ZZtoLep
104 prop_binSR_muon2017_bin15
105 prop_binCRTT_muon2016M_bin0
106 prop_binSR_muon2016M_bin10
107 prop_binSR_electron2017_bin9
108 prop_binCROS_muon2018_bin4
109 ISR_TVX
110 prop_binCROS_muon2018_bin5
111 prop_binCRF_muon2018_bin6
112 prop_binSR_electron2016M_bin9
113 prop_binSR_electron2017_bin3
114 prop_binCROS_electron2018_bin6
115 prop_binCRF_muon2018_bin3
116 prop_binSR_electron2017_bin12
117 prop_binCRF_muon2018_bin5
118 prop_binSR_electron2017_bin10
119 prop_binCRTT_muon2018_bin2
120 prop_binSR_electron2017_bin17
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

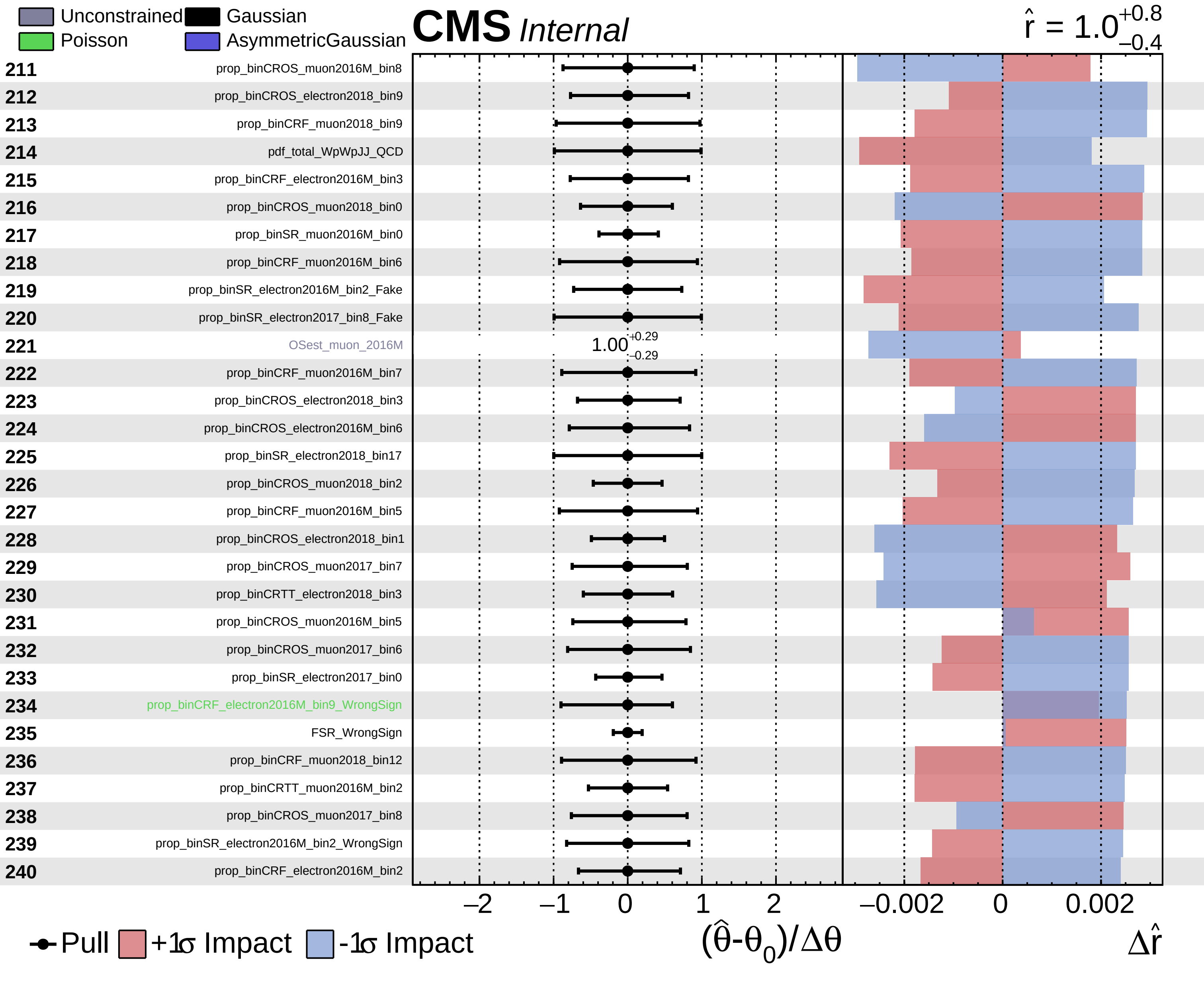


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
121 prop_binCRF_electron2016M_bin0
122 prop_binSR_electron2017_bin16
123 prop_binSR_muon2017_bin14_WrongSign
124 prop_binCRF_muon2018_bin1
125 prop_binCROS_electron2016M_bin1
126 VBS_2018
127 prop_binCRF_electron2017_bin0
128 prop_binCRF_muon2017_bin2
129 prop_binSR_electron2016M_bin5
130 prop_binSR_electron2017_bin11
131 prop_binCROS_muon2016M_bin0
132 VBS_2017
133 prop_binSR_muon2018_bin16
134 ISR_ZZtoLep
135 prop_binSR_electron2016M_bin3
136 prop_binCROS_electron2017_bin4
137 prop_binSR_electron2017_bin1
138 prop_binCROS_muon2016M_bin2
139 prop_binSR_electron2018_bin1
140 prop_binCROS_electron2018_bin7
141 prop_binCRF_muon2017_bin1
142 prop_binCROS_electron2018_bin2
143 prop_binSR_electron2016M_bin15
144 OSest_electron_2018
$1.00^{+0.32}_{-0.32}$
145 prop_binSR_electron2016M_bin4
146 prop_binSR_electron2018_bin12
147 prop_binSR_electron2018_bin15
148 prop_binSR_electron2016M_bin6
149 prop_binCRTT_electron2018_bin2
150 OSest_electron_2017
$1.00^{+0.29}_{-0.29}$
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

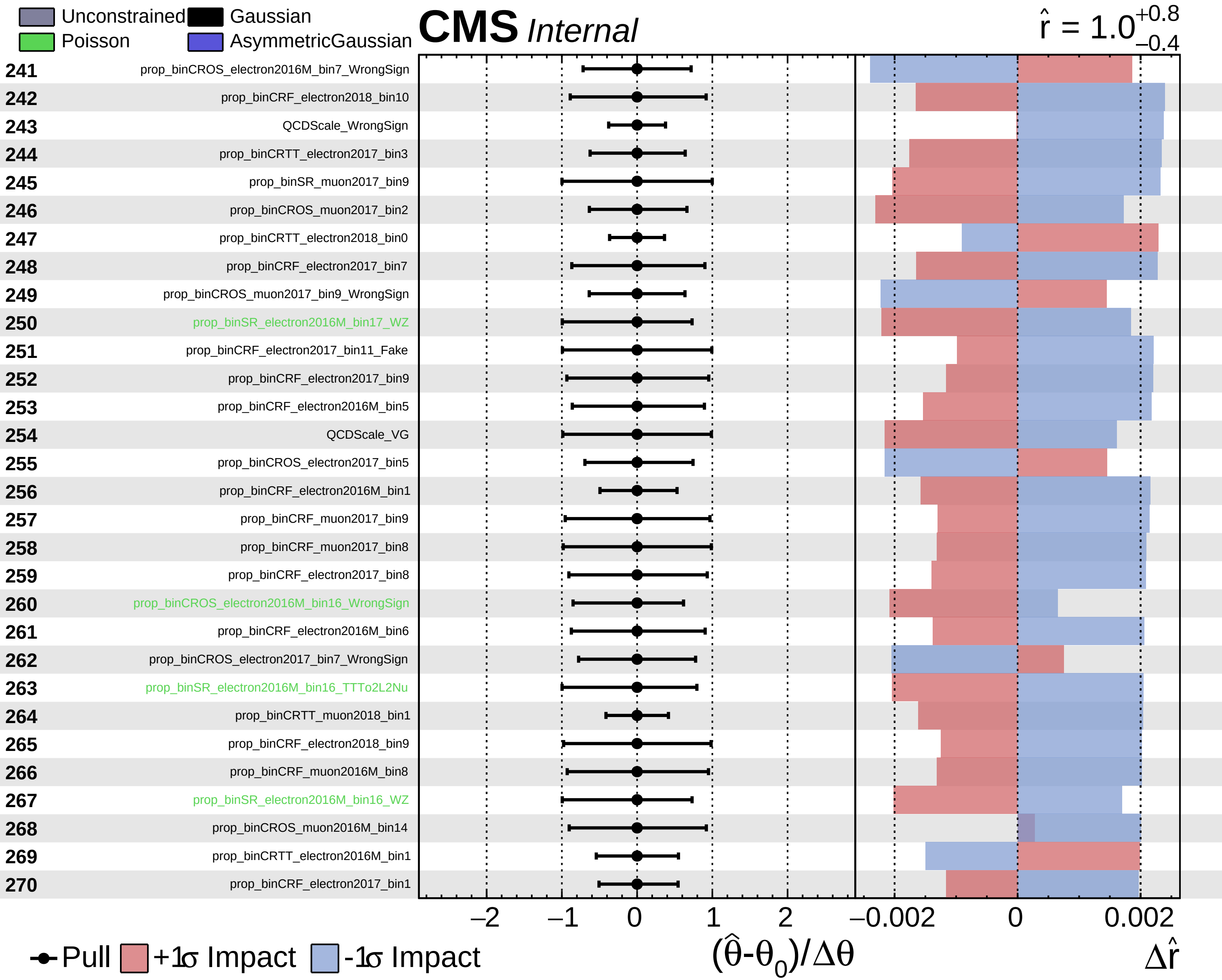

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
151 prop_binSR_muon2016M_bin15
152 metUnclust
153 prop_binSR_electron2016M_bin10
154 prop_binCROS_electron2018_bin11_WrongSign
155 prop_binCRF_muon2016M_bin1
156 TES
157 prop_binSR_electron2016M_bin8
158 prop_binCRF_muon2017_bin3
159 prop_binCRF_muon2017_bin4
160 prop_binSR_muon2016M_bin16
161 prop_binCROS_electron2017_bin0
162 prop_binCRF_muon2018_bin7
163 prop_binCRTT_muon2016M_bin1
164 QCDScale_Triboson
165 prop_binSR_muon2016M_bin17
166 prop_binSR_electron2016M_bin1
167 prop_binCRF_electron2018_bin2
168 prop_binCRF_electron2017_bin4
169 FSR_VG
170 prop_binCRF_muon2016M_bin4
171 prop_binCROS_electron2017_bin9
172 OSest_muon_2017
$1.00^{+0.23}_{-0.23}$
173 prop_binCRF_electron2018_bin4
174 prop_binSR_electron2016M_bin17_Fake
175 prop_binCRF_electron2017_bin2
176 prop_binCRF_electron2018_bin5
177 prop_binCRF_muon2018_bin8
178 prop_binCRF_muon2016M_bin3
179 prop_binCRF_electron2018_bin6
180 prop_binCROS_electron2018_bin0
−2
−1
0
1
2
−0.004
−0.002
0
0.002
0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

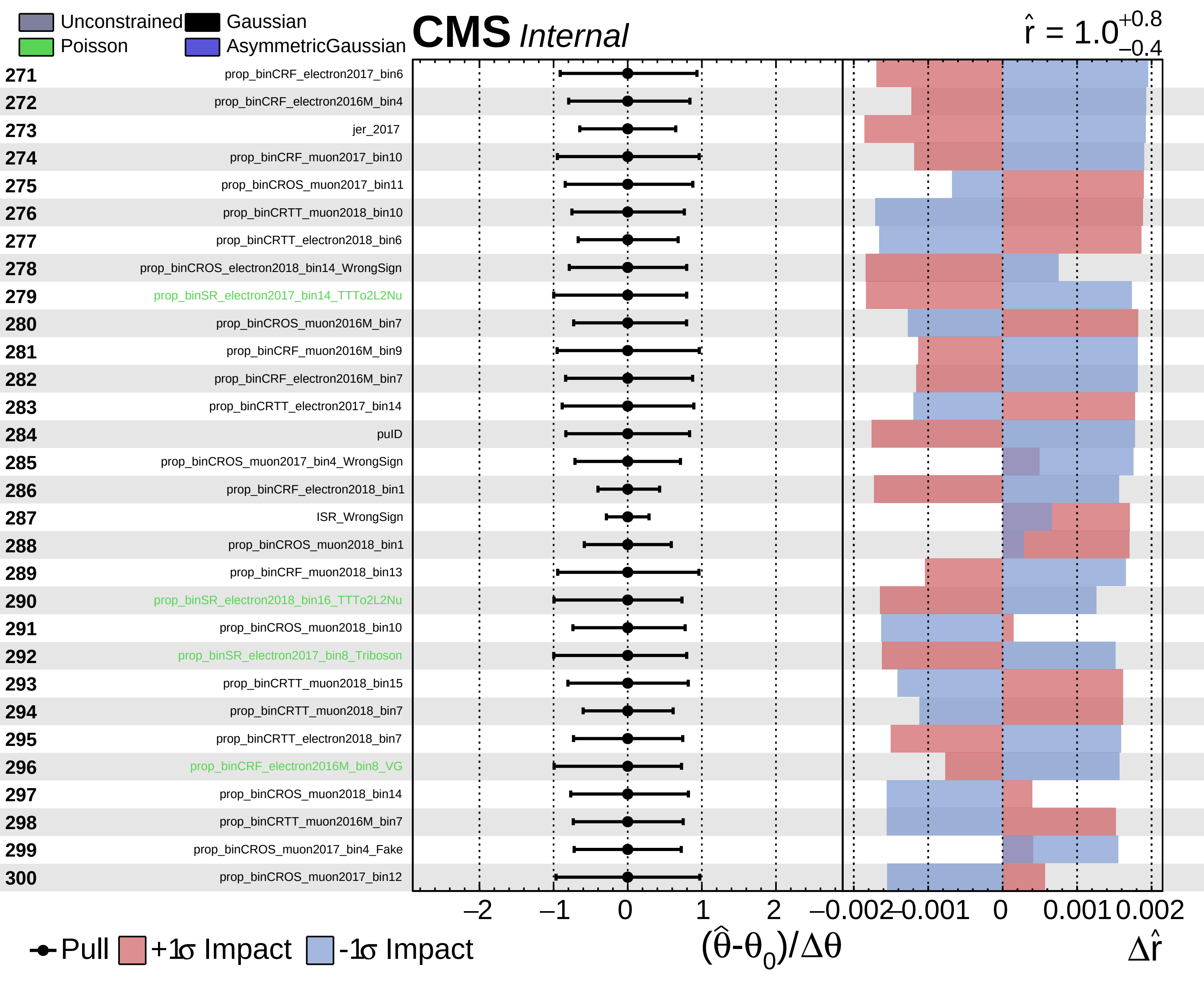


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
181 jer_2016M
182 prop_binCRTT_electron2017_bin5
183 prop_binCRF_electron2018_bin3
184 prop_binCROS_electron2017_bin3
185 prop_binCRF_muon2017_bin6
186 prop_binCRF_muon2017_bin0
187 prop_binCRF_muon2018_bin10
188 prop_binCROS_muon2018_bin9_WrongSign
189 prop_binCRF_muon2017_bin11_WZ
190 prop_binSR_muon2017_bin0
191 prop_binCROS_electron2017_bin6
192 VBS_2016M
193 prop_binSR_muon2016M_bin14_WZ
194 prop_binCRTT_muon2017_bin1
195 prop_binCRF_electron2018_bin8
196 prop_binCRF_muon2017_bin5
197 prop_binCROS_muon2018_bin3
198 prop_binCRTT_electron2016M_bin0
199 prop_binCRF_muon2016M_bin0
200 prop_binCROS_electron2018_bin12
201 prop_binCRF_electron2018_bin7
202 prop_binCRTT_muon2017_bin0
203 prop_binCROS_electron2016M_bin9_WrongSign
204 prop_binCRF_electron2017_bin5
205 prop_binCROS_muon2018_bin6
206 ISR_VG
207 prop_binCROS_muon2017_bin5
208 prop_binCRF_muon2017_bin7
209 prop_binSR_muon2018_bin0
210 prop_binSR_electron2016M_bin13
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

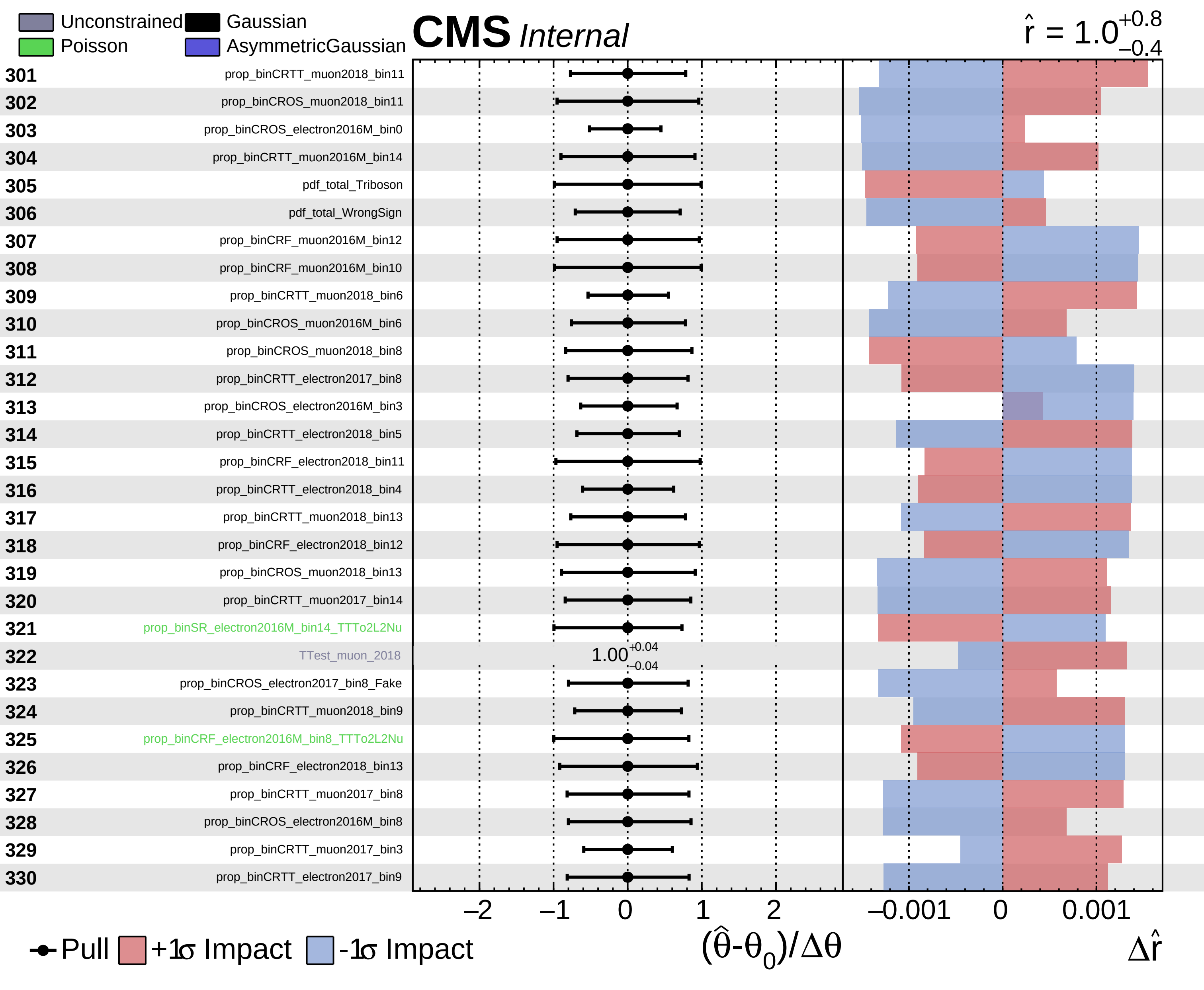


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
211 prop_binCROS_muon2016M_bin8
212 prop_binCROS_electron2018_bin9
213 prop_binCRF_muon2018_bin9
214 pdf_total_WpWpJJ_QCD
215 prop_binCRF_electron2016M_bin3
216 prop_binCROS_muon2018_bin0
217 prop_binSR_muon2016M_bin0
218 prop_binCRF_muon2016M_bin6
219 prop_binSR_electron2016M_bin2_Fake
220 prop_binSR_electron2017_bin8_Fake
221 OSest_muon_2016M
$1.00^{+0.29}_{-0.29}$
222 prop_binCRF_muon2016M_bin7
223 prop_binCROS_electron2018_bin3
224 prop_binCROS_electron2016M_bin6
225 prop_binSR_electron2018_bin17
226 prop_binCROS_muon2018_bin2
227 prop_binCRF_muon2016M_bin5
228 prop_binCROS_electron2018_bin1
229 prop_binCROS_muon2017_bin7
230 prop_binCRTT_electron2018_bin3
231 prop_binCROS_muon2016M_bin5
232 prop_binCROS_muon2017_bin6
233 prop_binSR_electron2017_bin0
234 prop_binCRF_electron2016M_bin9_WrongSign
235 FSR_WrongSign
236 prop_binCRF_muon2018_bin12
237 prop_binCRTT_muon2016M_bin2
238 prop_binCROS_muon2017_bin8
239 prop_binSR_electron2016M_bin2_WrongSign
240 prop_binCRF_electron2016M_bin2
−2 −1 0 1 2
−0.002 0 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

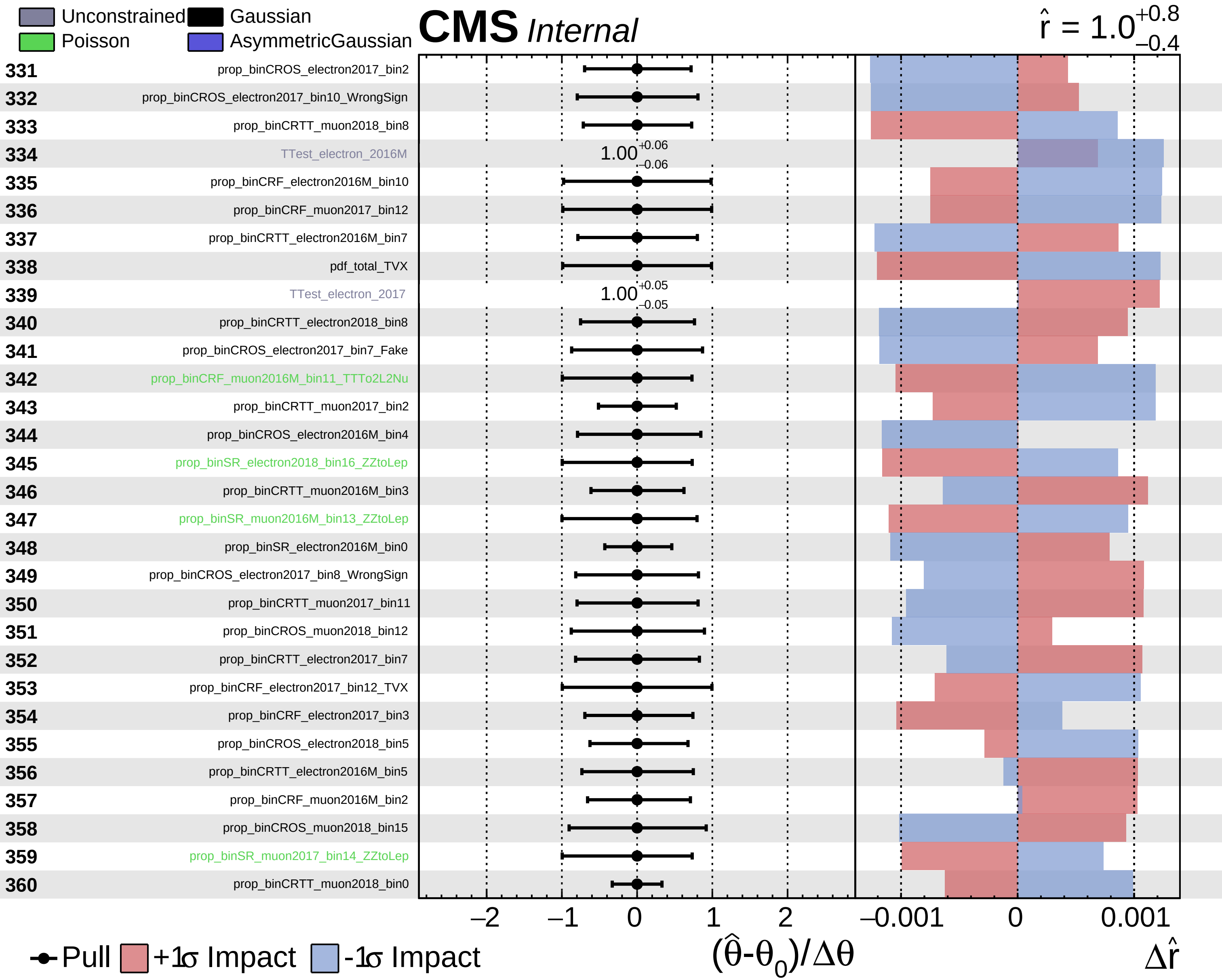

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
241 prop_binCROS_electron2016M_bin7_WrongSign
242 prop_binCRF_electron2018_bin10
243 QCDScale_WrongSign
244 prop_binCRTT_electron2017_bin3
245 prop_binSR_muon2017_bin9
246 prop_binCROS_muon2017_bin2
247 prop_binCRTT_electron2018_bin0
248 prop_binCRF_electron2017_bin7
249 prop_binCROS_muon2017_bin9_WrongSign
250 prop_binSR_electron2016M_bin17_WZ
251 prop_binCRF_electron2017_bin11_Fake
252 prop_binCRF_electron2017_bin9
253 prop_binCRF_electron2016M_bin5
254 QCDScale_VG
255 prop_binCROS_electron2017_bin5
256 prop_binCRF_electron2016M_bin1
257 prop_binCRF_muon2017_bin9
258 prop_binCRF_muon2017_bin8
259 prop_binCRF_electron2017_bin8
260 prop_binCROS_electron2016M_bin16_WrongSign
261 prop_binCRF_electron2016M_bin6
262 prop_binCROS_electron2017_bin7_WrongSign
263 prop_binSR_electron2016M_bin16_TTTo2L2Nu
264 prop_binCRTT_muon2018_bin1
265 prop_binCRF_electron2018_bin9
266 prop_binCRF_muon2016M_bin8
267 prop_binSR_electron2016M_bin16_WZ
268 prop_binCROS_muon2016M_bin14
269 prop_binCRTT_electron2016M_bin1
270 prop_binCRF_electron2017_bin1
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



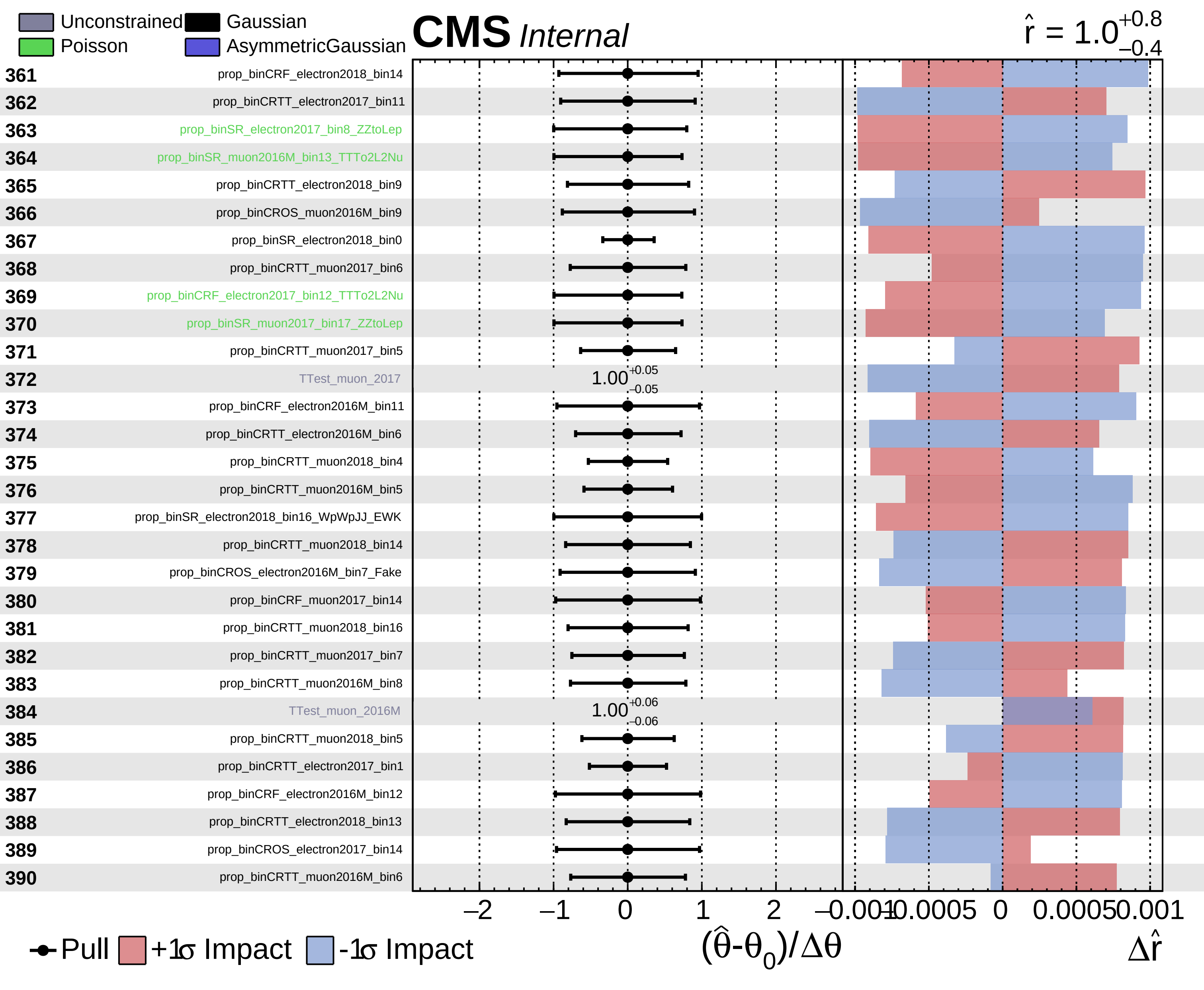
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
271 prop_binCRF_electron2017_bin6
272 prop_binCRF_electron2016M_bin4
273 jer_2017
274 prop_binCRF_muon2017_bin10
275 prop_binCROS_muon2017_bin11
276 prop_binCRTT_muon2018_bin10
277 prop_binCRTT_electron2018_bin6
278 prop_binCROS_electron2018_bin14_WrongSign
279 prop_binSR_electron2017_bin14_TTTo2L2Nu
280 prop_binCROS_muon2016M_bin7
281 prop_binCRF_muon2016M_bin9
282 prop_binCRF_electron2016M_bin7
283 prop_binCRTT_electron2017_bin14
284 puID
285 prop_binCROS_muon2017_bin4_WrongSign
286 prop_binCRF_electron2018_bin1
287 ISR_WrongSign
288 prop_binCROS_muon2018_bin1
289 prop_binCRF_muon2018_bin13
290 prop_binSR_electron2018_bin16_TTTo2L2Nu
291 prop_binCROS_muon2018_bin10
292 prop_binSR_electron2017_bin8_Triboson
293 prop_binCRTT_muon2018_bin15
294 prop_binCRTT_muon2018_bin7
295 prop_binCRTT_electron2018_bin7
296 prop_binCRF_electron2016M_bin8_VG
297 prop_binCROS_muon2018_bin14
298 prop_binCRTT_muon2016M_bin7
299 prop_binCROS_muon2017_bin4_Fake
300 prop_binCROS_muon2017_bin12
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



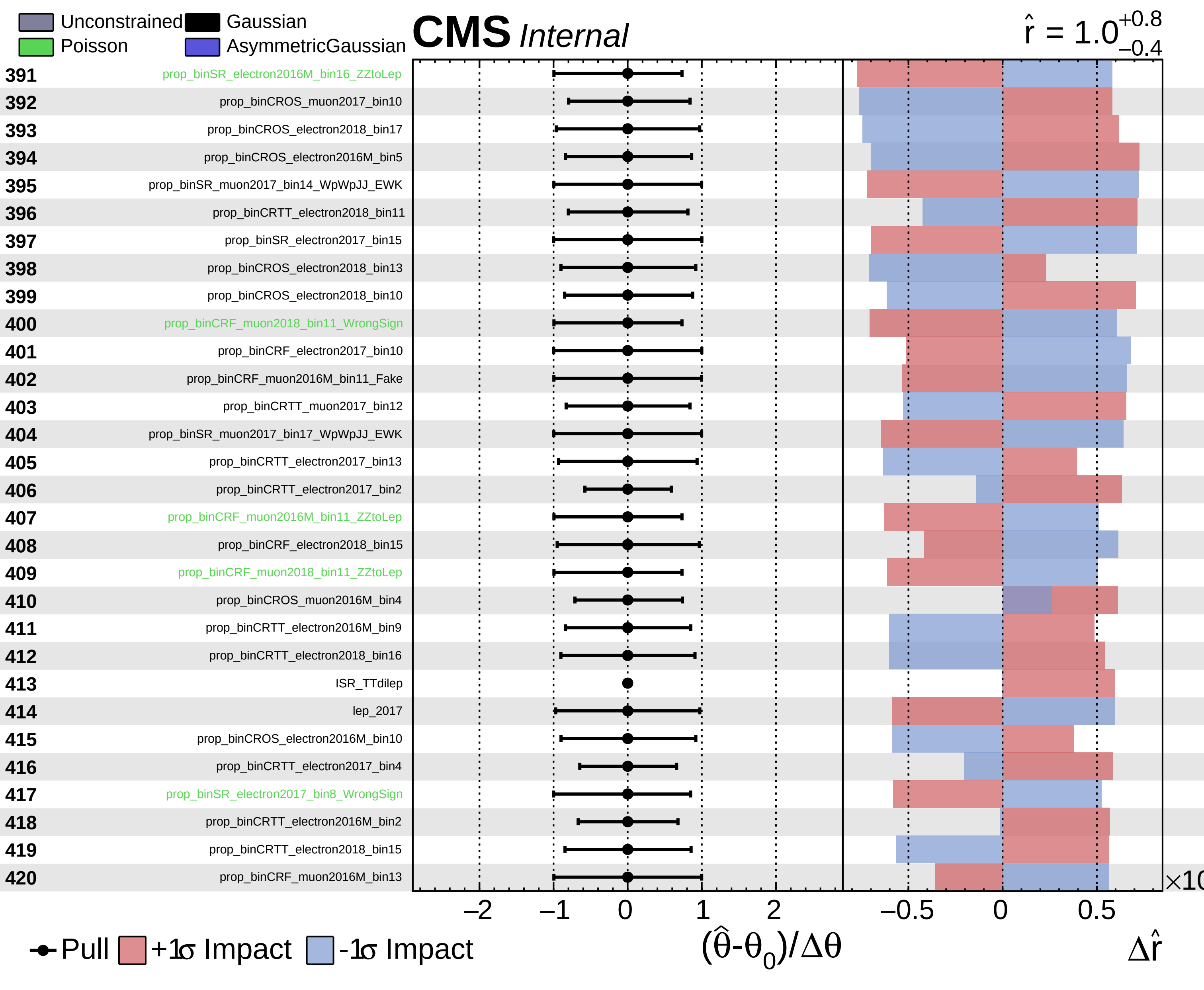
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
301 prop_binCRTT_muon2018_bin11
302 prop_binCROS_muon2018_bin11
303 prop_binCROS_electron2016M_bin0
304 prop_binCRTT_muon2016M_bin14
305 pdf_total_Triboson
306 pdf_total_WrongSign
307 prop_binCRF_muon2016M_bin12
308 prop_binCRF_muon2016M_bin10
309 prop_binCRTT_muon2018_bin6
310 prop_binCROS_muon2016M_bin6
311 prop_binCROS_muon2018_bin8
312 prop_binCRTT_electron2017_bin8
313 prop_binCROS_electron2016M_bin3
314 prop_binCRTT_electron2018_bin5
315 prop_binCRF_electron2018_bin11
316 prop_binCRTT_electron2018_bin4
317 prop_binCRTT_muon2018_bin13
318 prop_binCRF_electron2018_bin12
319 prop_binCROS_muon2018_bin13
320 prop_binCRTT_muon2017_bin14
321 prop_binSR_electron2016M_bin14_TTTo2L2Nu
322 TTest_muon_2018
$1.00^{+0.04}_{-0.04}$
323 prop_binCROS_electron2017_bin8_Fake
324 prop_binCRTT_muon2018_bin9
325 prop_binCRF_electron2016M_bin8_TTTo2L2Nu
326 prop_binCRF_electron2018_bin13
327 prop_binCRTT_muon2017_bin8
328 prop_binCROS_electron2016M_bin8
329 prop_binCRTT_muon2017_bin3
330 prop_binCRTT_electron2017_bin9
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



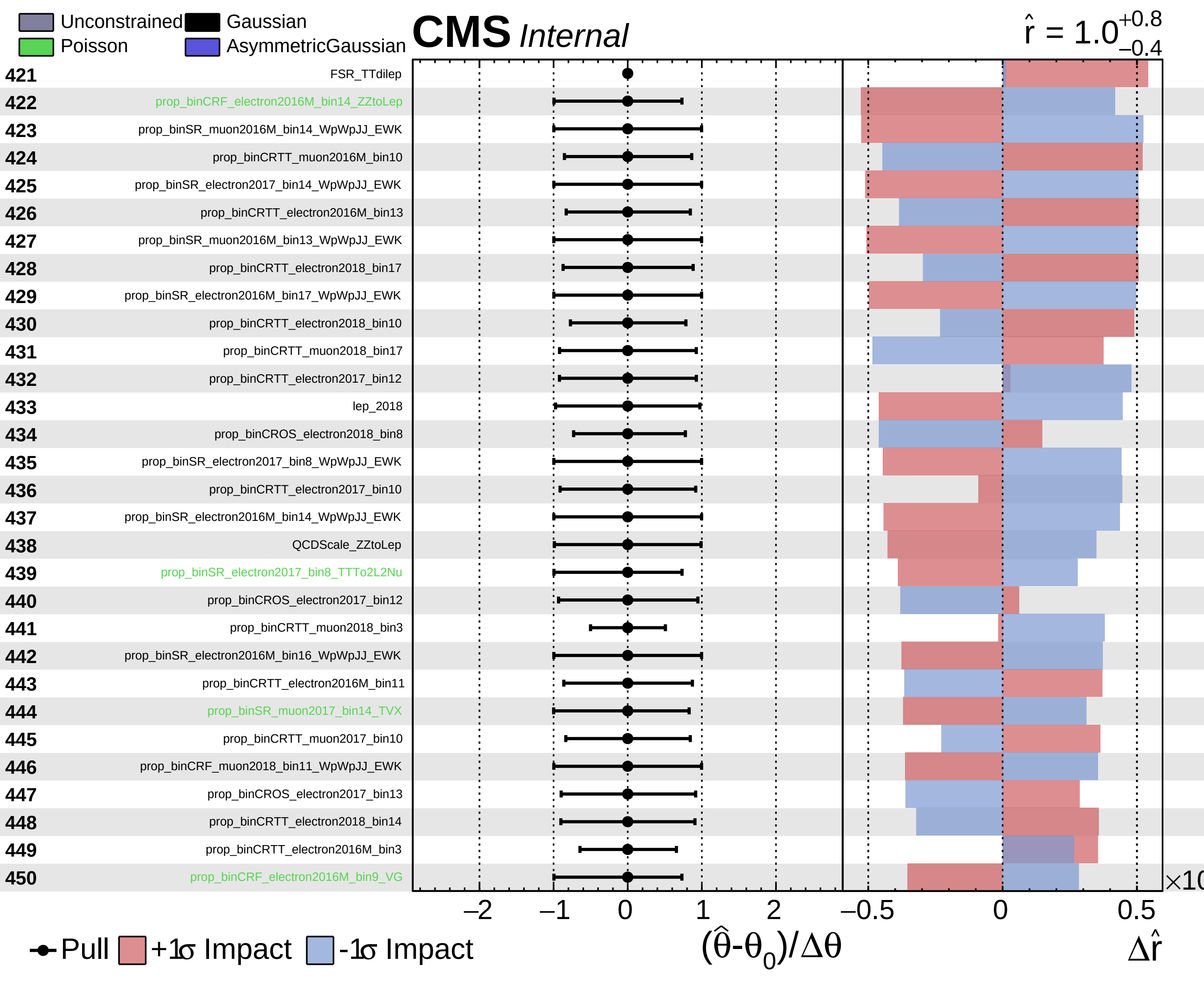
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.8 −0.4
331 prop_binCROS_electron2017_bin2
332 prop_binCROS_electron2017_bin10_WrongSign
333 prop_binCRTT_muon2018_bin8
334 TTest_electron_2016M
1.00 +0.06 −0.06
335 prop_binCRF_electron2016M_bin10
336 prop_binCRF_muon2017_bin12
337 prop_binCRTT_electron2016M_bin7
338 pdf_total_TVX
339 TTest_electron_2017
1.00 +0.05 −0.05
340 prop_binCRTT_electron2018_bin8
341 prop_binCROS_electron2017_bin7_Fake
342 prop_binCRF_muon2016M_bin11_TTTo2L2Nu
343 prop_binCRTT_muon2017_bin2
344 prop_binCROS_electron2016M_bin4
345 prop_binSR_electron2018_bin16_ZZtoLep
346 prop_binCRTT_muon2016M_bin3
347 prop_binSR_muon2016M_bin13_ZZtoLep
348 prop_binSR_electron2016M_bin0
349 prop_binCROS_electron2017_bin8_WrongSign
350 prop_binCRTT_muon2017_bin11
351 prop_binCROS_muon2018_bin12
352 prop_binCRTT_electron2017_bin7
353 prop_binCRF_electron2017_bin12_TVX
354 prop_binCRF_electron2017_bin3
355 prop_binCROS_electron2018_bin5
356 prop_binCRTT_electron2016M_bin5
357 prop_binCRF_muon2016M_bin2
358 prop_binCROS_muon2018_bin15
359 prop_binSR_muon2017_bin14_ZZtoLep
360 prop_binCRTT_muon2018_bin0
−2 −1 0 1 2
−0.001 0 0.001
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



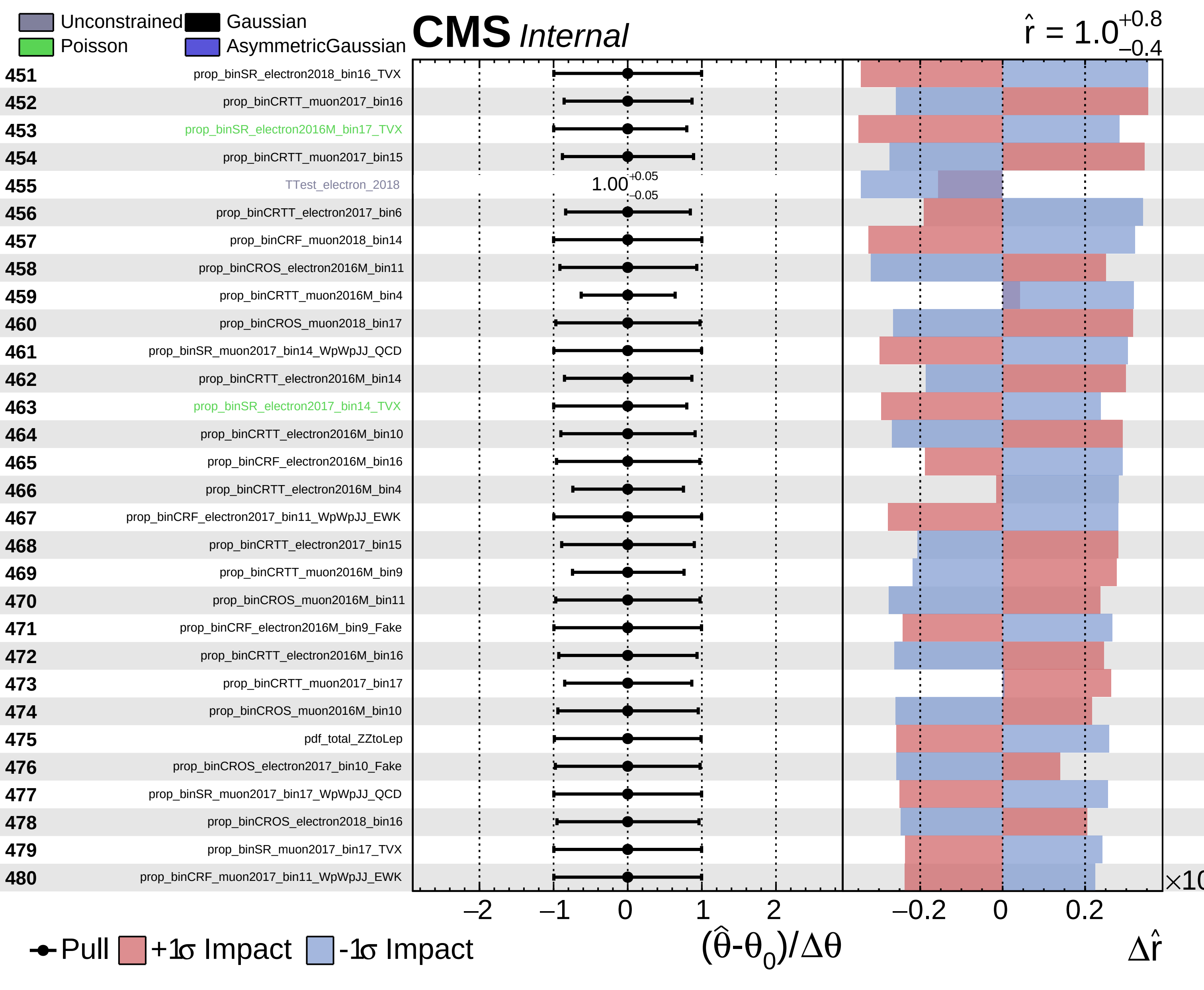
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.8 −0.4
361 prop_binCRF_electron2018_bin14
362 prop_binCRTT_electron2017_bin11
363 prop_binSR_electron2017_bin8_ZZtoLep
364 prop_binSR_muon2016M_bin13_TTTo2L2Nu
365 prop_binCRTT_electron2018_bin9
366 prop_binCROS_muon2016M_bin9
367 prop_binSR_electron2018_bin0
368 prop_binCRTT_muon2017_bin6
369 prop_binCRF_electron2017_bin12_TTTo2L2Nu
370 prop_binSR_muon2017_bin17_ZZtoLep
371 prop_binCRTT_muon2017_bin5
372 TTest_muon_2017
1.00 +0.05 −0.05
373 prop_binCRF_electron2016M_bin11
374 prop_binCRTT_electron2016M_bin6
375 prop_binCRTT_muon2018_bin4
376 prop_binCRTT_muon2016M_bin5
377 prop_binSR_electron2018_bin16_WpWpJJ_EWK
378 prop_binCRTT_muon2018_bin14
379 prop_binCROS_electron2016M_bin7_Fake
380 prop_binCRF_muon2017_bin14
381 prop_binCRTT_muon2018_bin16
382 prop_binCRTT_muon2017_bin7
383 prop_binCRTT_muon2016M_bin8
384 TTest_muon_2016M
1.00 +0.06 −0.06
385 prop_binCRTT_muon2018_bin5
386 prop_binCRTT_electron2017_bin1
387 prop_binCRF_electron2016M_bin12
388 prop_binCRTT_electron2018_bin13
389 prop_binCROS_electron2017_bin14
390 prop_binCRTT_muon2016M_bin6
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



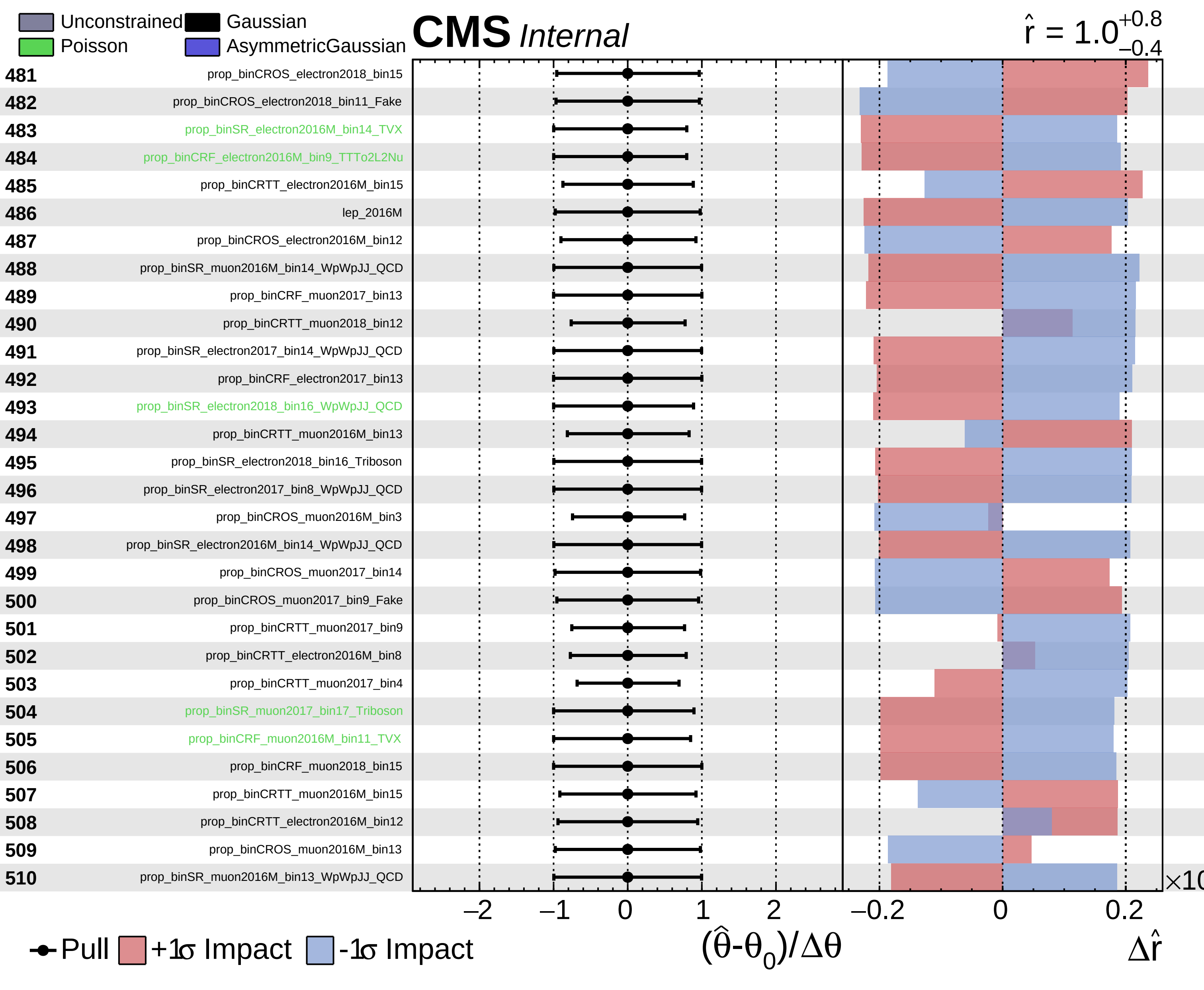
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
391 prop_binSR_electron2016M_bin16_ZZtoLep
392 prop_binCROS_muon2017_bin10
393 prop_binCROS_electron2018_bin17
394 prop_binCROS_electron2016M_bin5
395 prop_binSR_muon2017_bin14_WpWpJJ_EWK
396 prop_binCRTT_electron2018_bin11
397 prop_binSR_electron2017_bin15
398 prop_binCROS_electron2018_bin13
399 prop_binCROS_electron2018_bin10
400 prop_binCRF_muon2018_bin11_WrongSign
401 prop_binCRF_electron2017_bin10
402 prop_binCRF_muon2016M_bin11_Fake
403 prop_binCRTT_muon2017_bin12
404 prop_binSR_muon2017_bin17_WpWpJJ_EWK
405 prop_binCRTT_electron2017_bin13
406 prop_binCRTT_electron2017_bin2
407 prop_binCRF_muon2016M_bin11_ZZtoLep
408 prop_binCRF_electron2018_bin15
409 prop_binCRF_muon2018_bin11_ZZtoLep
410 prop_binCROS_muon2016M_bin4
411 prop_binCRTT_electron2016M_bin9
412 prop_binCRTT_electron2018_bin16
413 ISR_TTdilep
414 lep_2017
415 prop_binCROS_electron2016M_bin10
416 prop_binCRTT_electron2017_bin4
417 prop_binSR_electron2017_bin8_WrongSign
418 prop_binCRTT_electron2016M_bin2
419 prop_binCRTT_electron2018_bin15
420 prop_binCRF_muon2016M_bin13
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

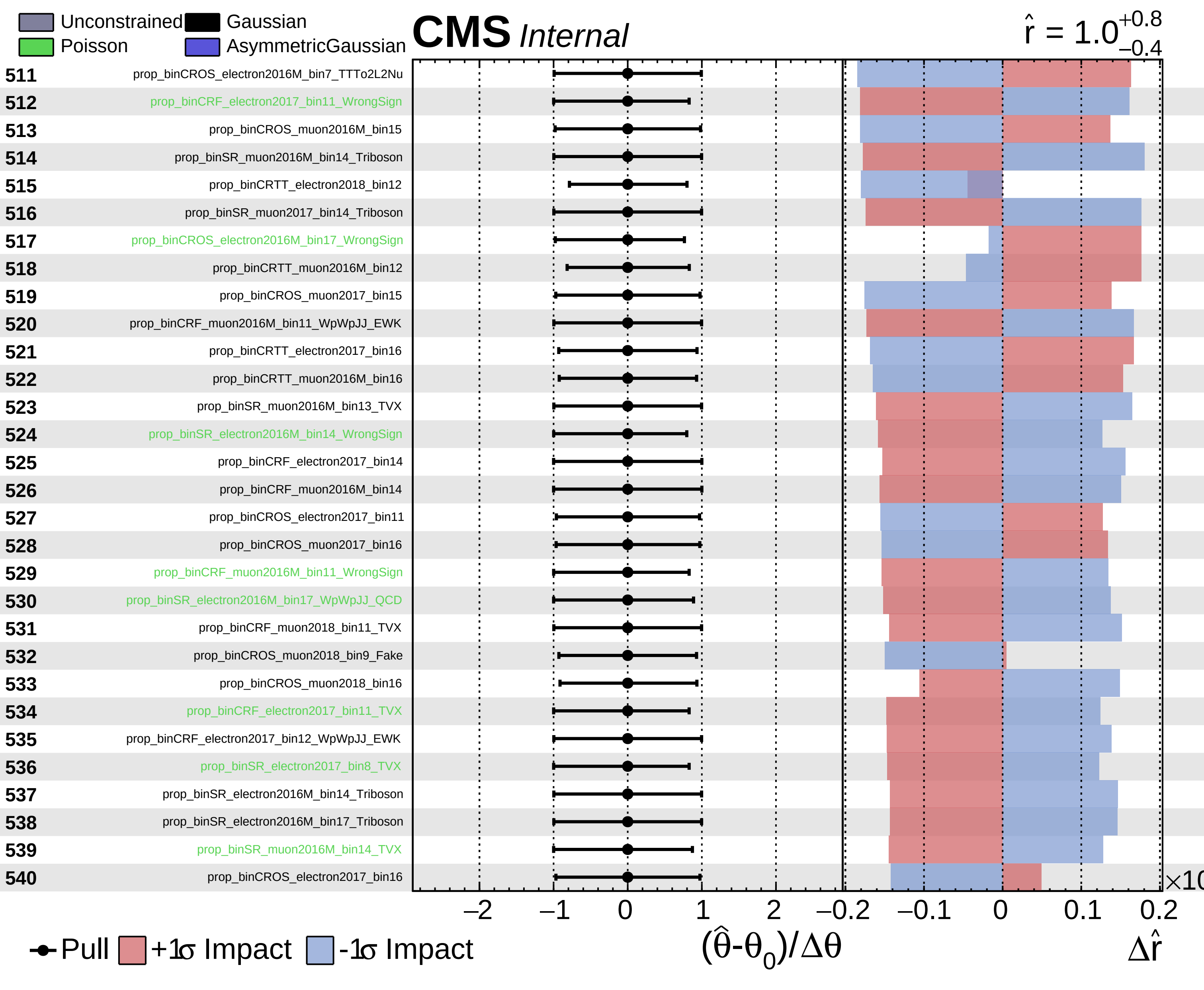


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
421 FSR_TTdilep
422 prop_binCRF_electron2016M_bin14_ZZtoLep
423 prop_binSR_muon2016M_bin14_WpWpJJ_EWK
424 prop_binCRTT_muon2016M_bin10
425 prop_binSR_electron2017_bin14_WpWpJJ_EWK
426 prop_binCRTT_electron2016M_bin13
427 prop_binSR_muon2016M_bin13_WpWpJJ_EWK
428 prop_binCRTT_electron2018_bin17
429 prop_binSR_electron2016M_bin17_WpWpJJ_EWK
430 prop_binCRTT_electron2018_bin10
431 prop_binCRTT_muon2018_bin17
432 prop_binCRTT_electron2017_bin12
433 lep_2018
434 prop_binCROS_electron2018_bin8
435 prop_binSR_electron2017_bin8_WpWpJJ_EWK
436 prop_binCRTT_electron2017_bin10
437 prop_binSR_electron2016M_bin14_WpWpJJ_EWK
438 QCDScale_ZZtoLep
439 prop_binSR_electron2017_bin8_TTTo2L2Nu
440 prop_binCROS_electron2017_bin12
441 prop_binCRTT_muon2018_bin3
442 prop_binSR_electron2016M_bin16_WpWpJJ_EWK
443 prop_binCRTT_electron2016M_bin11
444 prop_binSR_muon2017_bin14_TVX
445 prop_binCRTT_muon2017_bin10
446 prop_binCRF_muon2018_bin11_WpWpJJ_EWK
447 prop_binCROS_electron2017_bin13
448 prop_binCRTT_electron2018_bin14
449 prop_binCRTT_electron2016M_bin3
450 prop_binCRF_electron2016M_bin9_VG
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

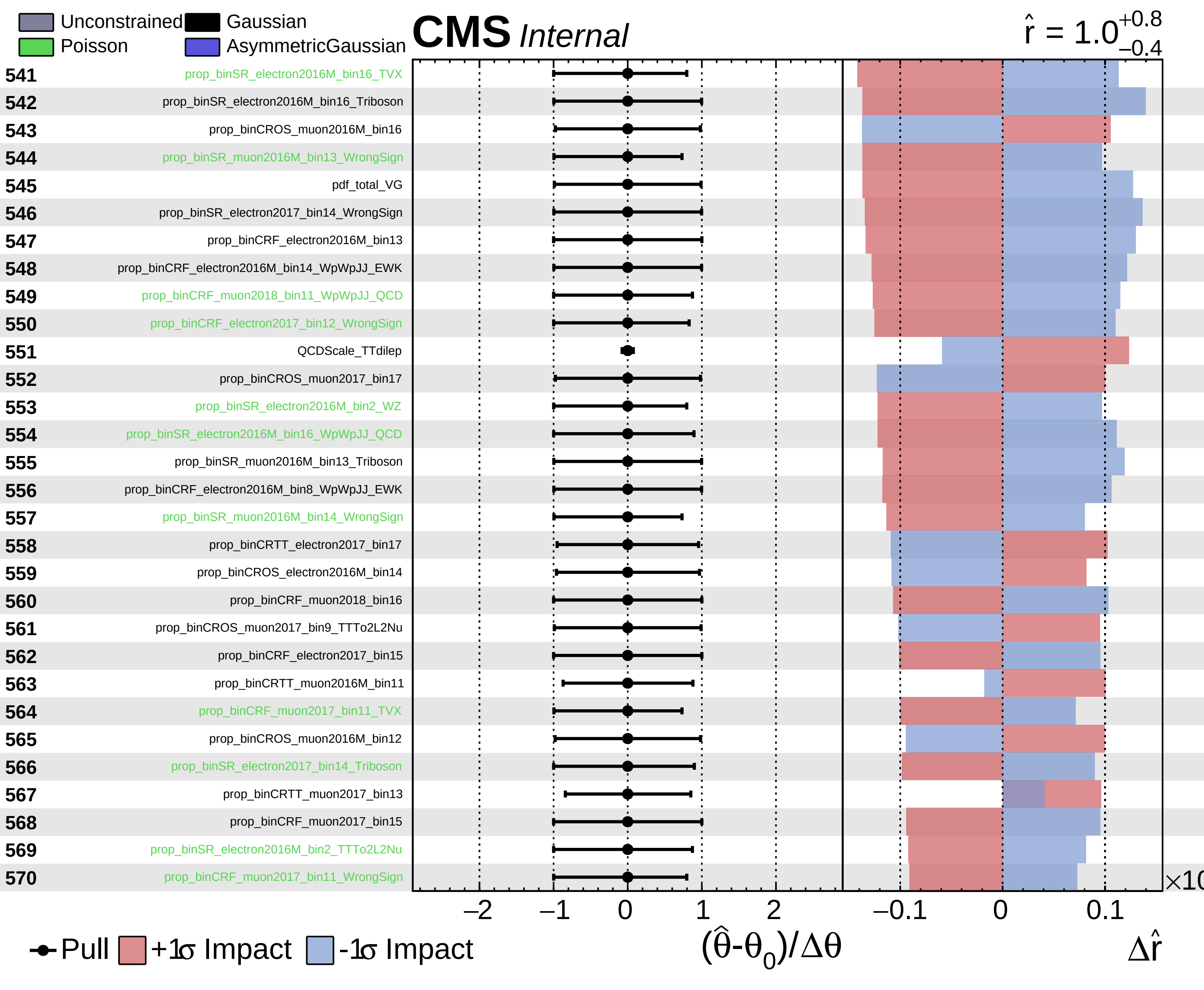


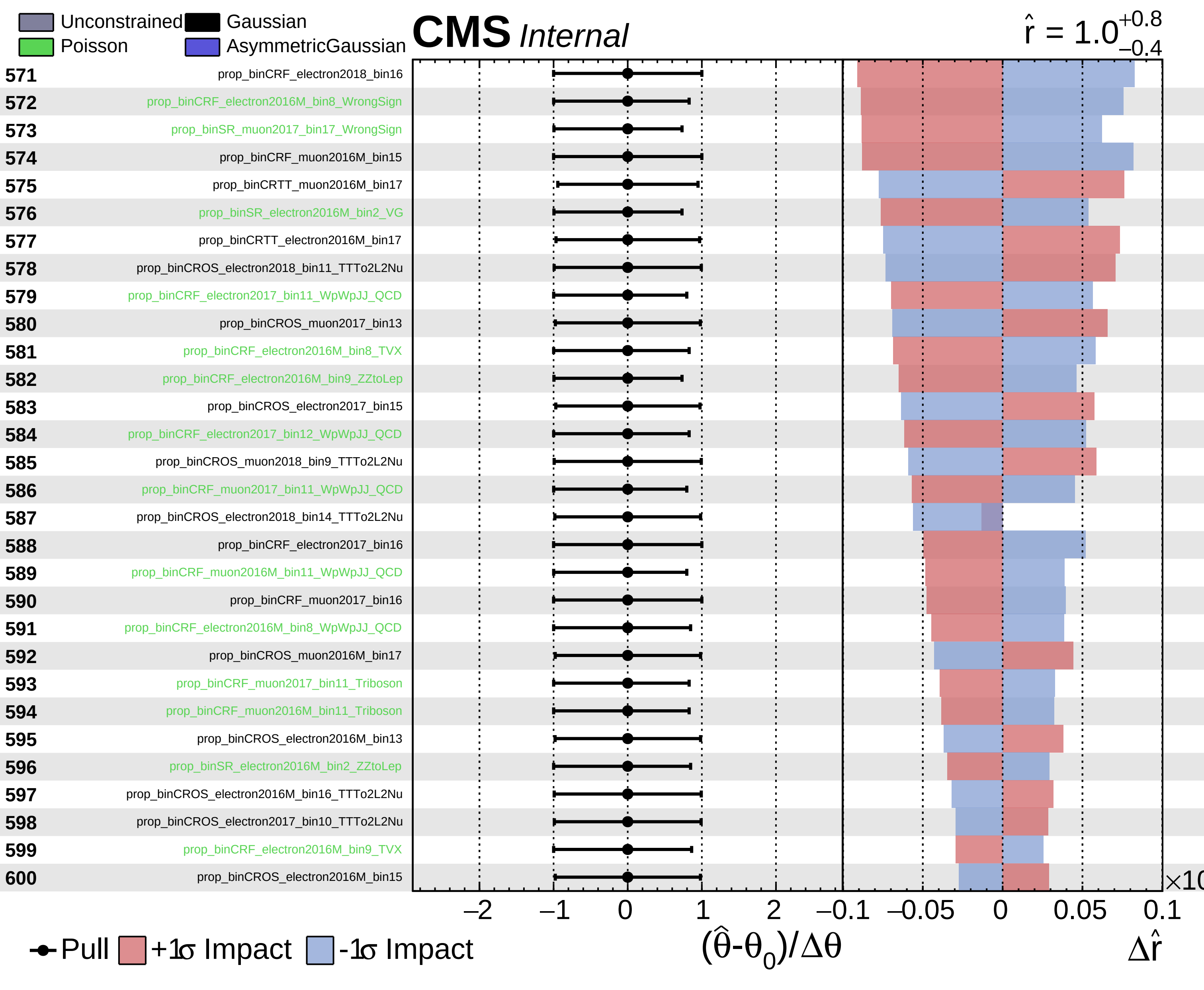
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.8 −0.4
451 prop_binSR_electron2018_bin16_TVX
452 prop_binCRTT_muon2017_bin16
453 prop_binSR_electron2016M_bin17_TVX
454 prop_binCRTT_muon2017_bin15
455 TTest_electron_2018
1.00 +0.05 −0.05
456 prop_binCRTT_electron2017_bin6
457 prop_binCRF_muon2018_bin14
458 prop_binCROS_electron2016M_bin11
459 prop_binCRTT_muon2016M_bin4
460 prop_binCROS_muon2018_bin17
461 prop_binSR_muon2017_bin14_WpWpJJ_QCD
462 prop_binCRTT_electron2016M_bin14
463 prop_binSR_electron2017_bin14_TVX
464 prop_binCRTT_electron2016M_bin10
465 prop_binCRF_electron2016M_bin16
466 prop_binCRTT_electron2016M_bin4
467 prop_binCRF_electron2017_bin11_WpWpJJ_EWK
468 prop_binCRTT_electron2017_bin15
469 prop_binCRTT_muon2016M_bin9
470 prop_binCROS_muon2016M_bin11
471 prop_binCRF_electron2016M_bin9_Fake
472 prop_binCRTT_electron2016M_bin16
473 prop_binCRTT_muon2017_bin17
474 prop_binCROS_muon2016M_bin10
475 pdf_total_ZZtoLep
476 prop_binCROS_electron2017_bin10_Fake
477 prop_binSR_muon2017_bin17_WpWpJJ_QCD
478 prop_binCROS_electron2018_bin16
479 prop_binSR_muon2017_bin17_TVX
480 prop_binCRF_muon2017_bin11_WpWpJJ_EWK
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
481 prop_binCROS_electron2018_bin15
482 prop_binCROS_electron2018_bin11_Fake
483 prop_binSR_electron2016M_bin14_TVX
484 prop_binCRF_electron2016M_bin9_TTTo2L2Nu
485 prop_binCRTT_electron2016M_bin15
486 lep_2016M
487 prop_binCROS_electron2016M_bin12
488 prop_binSR_muon2016M_bin14_WpWpJJ_QCD
489 prop_binCRF_muon2017_bin13
490 prop_binCRTT_muon2018_bin12
491 prop_binSR_electron2017_bin14_WpWpJJ_QCD
492 prop_binCRF_electron2017_bin13
493 prop_binSR_electron2018_bin16_WpWpJJ_QCD
494 prop_binCRTT_muon2016M_bin13
495 prop_binSR_electron2018_bin16_Triboson
496 prop_binSR_electron2017_bin8_WpWpJJ_QCD
497 prop_binCROS_muon2016M_bin3
498 prop_binSR_electron2016M_bin14_WpWpJJ_QCD
499 prop_binCROS_muon2017_bin14
500 prop_binCROS_muon2017_bin9_Fake
501 prop_binCRTT_muon2017_bin9
502 prop_binCRTT_electron2016M_bin8
503 prop_binCRTT_muon2017_bin4
504 prop_binSR_muon2017_bin17_Triboson
505 prop_binCRF_muon2016M_bin11_TVX
506 prop_binCRF_muon2018_bin15
507 prop_binCRTT_muon2016M_bin15
508 prop_binCRTT_electron2016M_bin12
509 prop_binCROS_muon2016M_bin13
510 prop_binSR_muon2016M_bin13_WpWpJJ_QCD
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

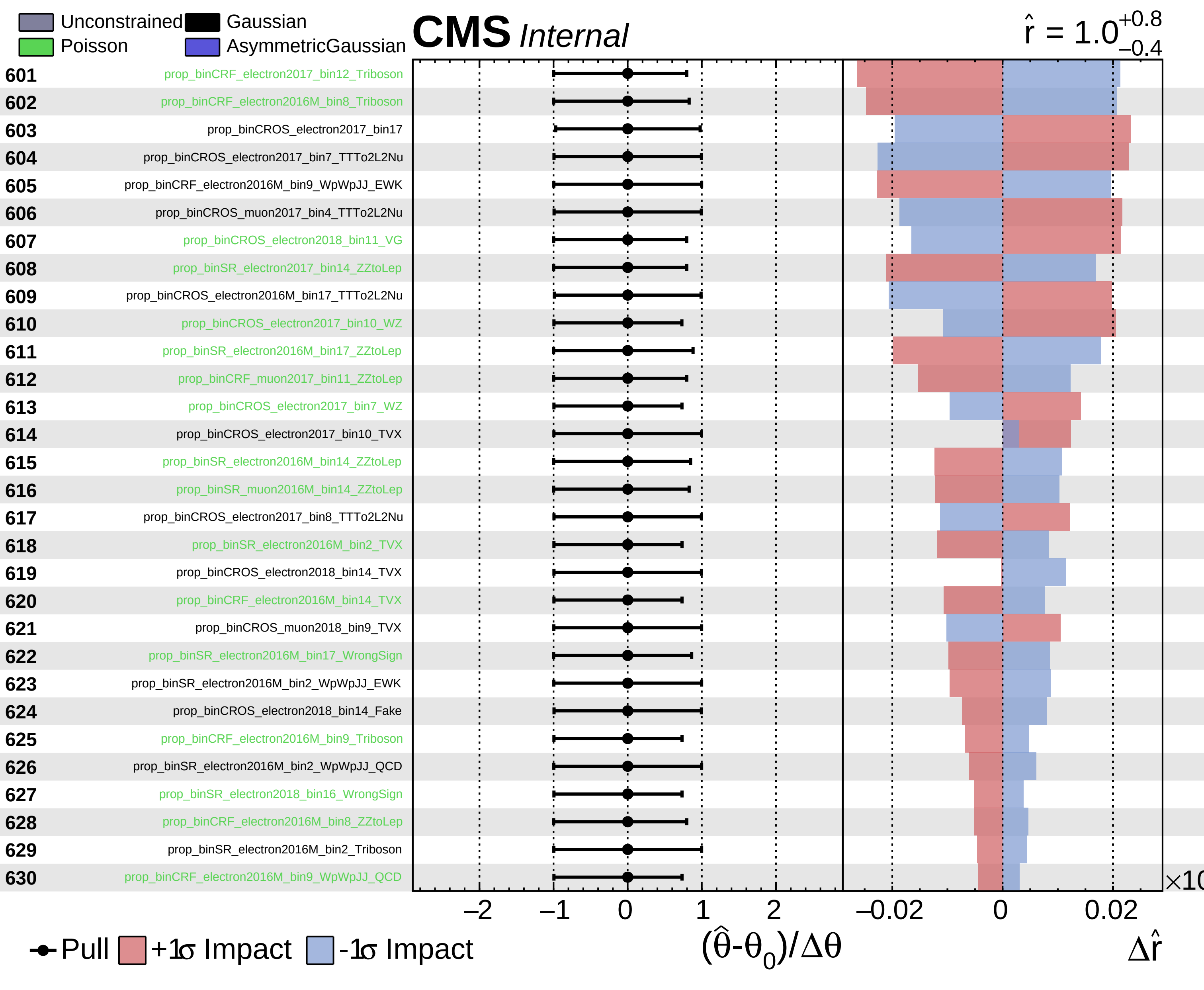


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
511 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
512 prop_binCRF_electron2017_bin11_WrongSign
513 prop_binCROS_muon2016M_bin15
514 prop_binSR_muon2016M_bin14_Triboson
515 prop_binCRTT_electron2018_bin12
516 prop_binSR_muon2017_bin14_Triboson
517 prop_binCROS_electron2016M_bin17_WrongSign
518 prop_binCRTT_muon2016M_bin12
519 prop_binCROS_muon2017_bin15
520 prop_binCRF_muon2016M_bin11_WpWpJJ_EWK
521 prop_binCRTT_electron2017_bin16
522 prop_binCRTT_muon2016M_bin16
523 prop_binSR_muon2016M_bin13_TVX
524 prop_binSR_electron2016M_bin14_WrongSign
525 prop_binCRF_electron2017_bin14
526 prop_binCRF_muon2016M_bin14
527 prop_binCROS_electron2017_bin11
528 prop_binCROS_muon2017_bin16
529 prop_binCRF_muon2016M_bin11_WrongSign
530 prop_binSR_electron2016M_bin17_WpWpJJ_QCD
531 prop_binCRF_muon2018_bin11_TVX
532 prop_binCROS_muon2018_bin9_Fake
533 prop_binCROS_muon2018_bin16
534 prop_binCRF_electron2017_bin11_TVX
535 prop_binCRF_electron2017_bin12_WpWpJJ_EWK
536 prop_binSR_electron2017_bin8_TVX
537 prop_binSR_electron2016M_bin14_Triboson
538 prop_binSR_electron2016M_bin17_Triboson
539 prop_binSR_muon2016M_bin14_TVX
540 prop_binCROS_electron2017_bin16
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

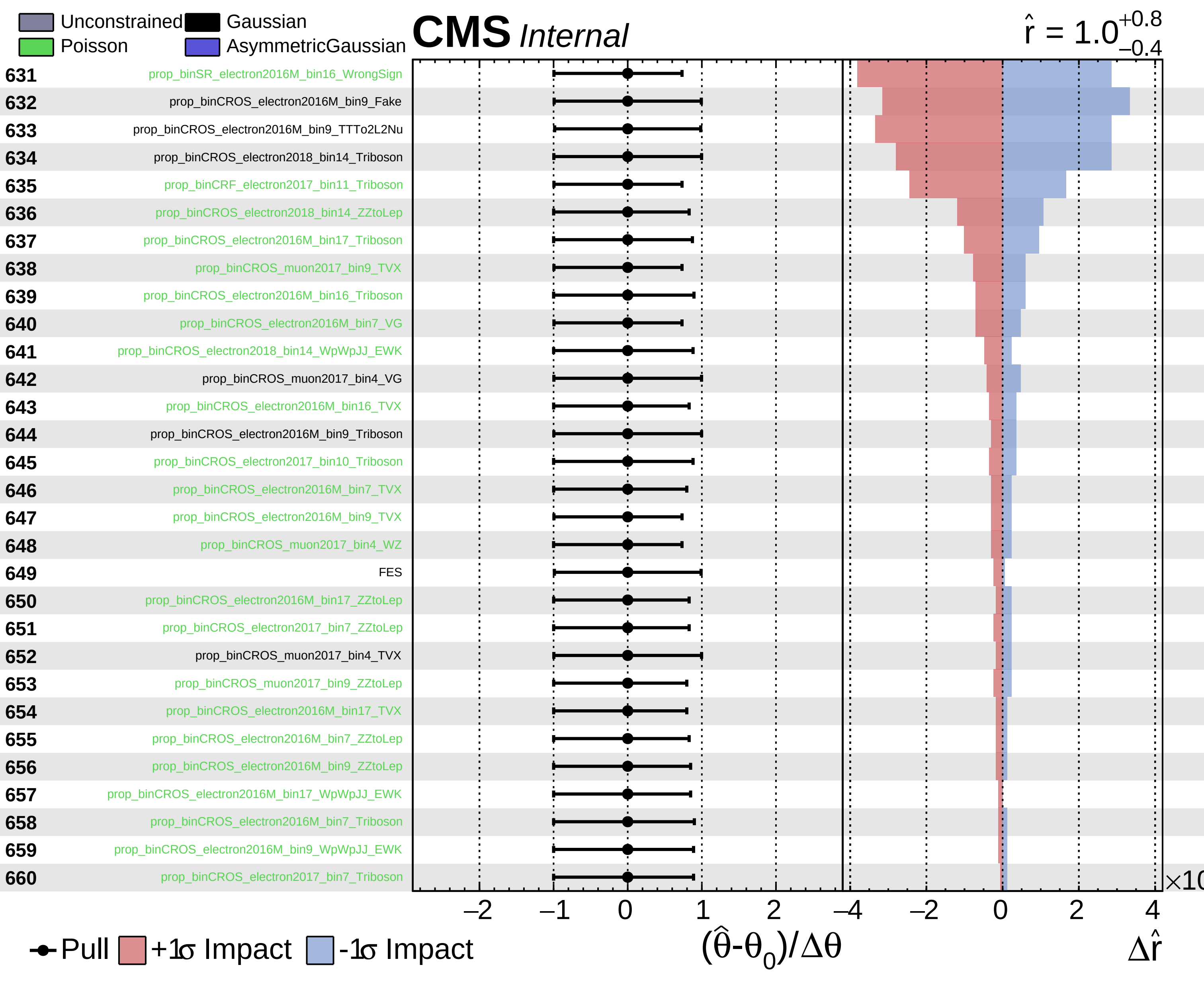

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
541 prop_binSR_electron2016M_bin16_TVX
542 prop_binSR_electron2016M_bin16_Triboson
543 prop_binCROS_muon2016M_bin16
544 prop_binSR_muon2016M_bin13_WrongSign
545 pdf_total_VG
546 prop_binSR_electron2017_bin14_WrongSign
547 prop_binCRF_electron2016M_bin13
548 prop_binCRF_electron2016M_bin14_WpWpJJ_EWK
549 prop_binCRF_muon2018_bin11_WpWpJJ_QCD
550 prop_binCRF_electron2017_bin12_WrongSign
551 QCDScale_TTdilep
552 prop_binCROS_muon2017_bin17
553 prop_binSR_electron2016M_bin2_WZ
554 prop_binSR_electron2016M_bin16_WpWpJJ_QCD
555 prop_binSR_muon2016M_bin13_Triboson
556 prop_binCRF_electron2016M_bin8_WpWpJJ_EWK
557 prop_binSR_muon2016M_bin14_WrongSign
558 prop_binCRTT_electron2017_bin17
559 prop_binCROS_electron2016M_bin14
560 prop_binCRF_muon2018_bin16
561 prop_binCROS_muon2017_bin9_TTTo2L2Nu
562 prop_binCRF_electron2017_bin15
563 prop_binCRTT_muon2016M_bin11
564 prop_binCRF_muon2017_bin11_TVX
565 prop_binCROS_muon2016M_bin12
566 prop_binSR_electron2017_bin14_Triboson
567 prop_binCRTT_muon2017_bin13
568 prop_binCRF_muon2017_bin15
569 prop_binSR_electron2016M_bin2_TTTo2L2Nu
570 prop_binCRF_muon2017_bin11_WrongSign
−2
−1
0
1
2
−0.1
0
0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

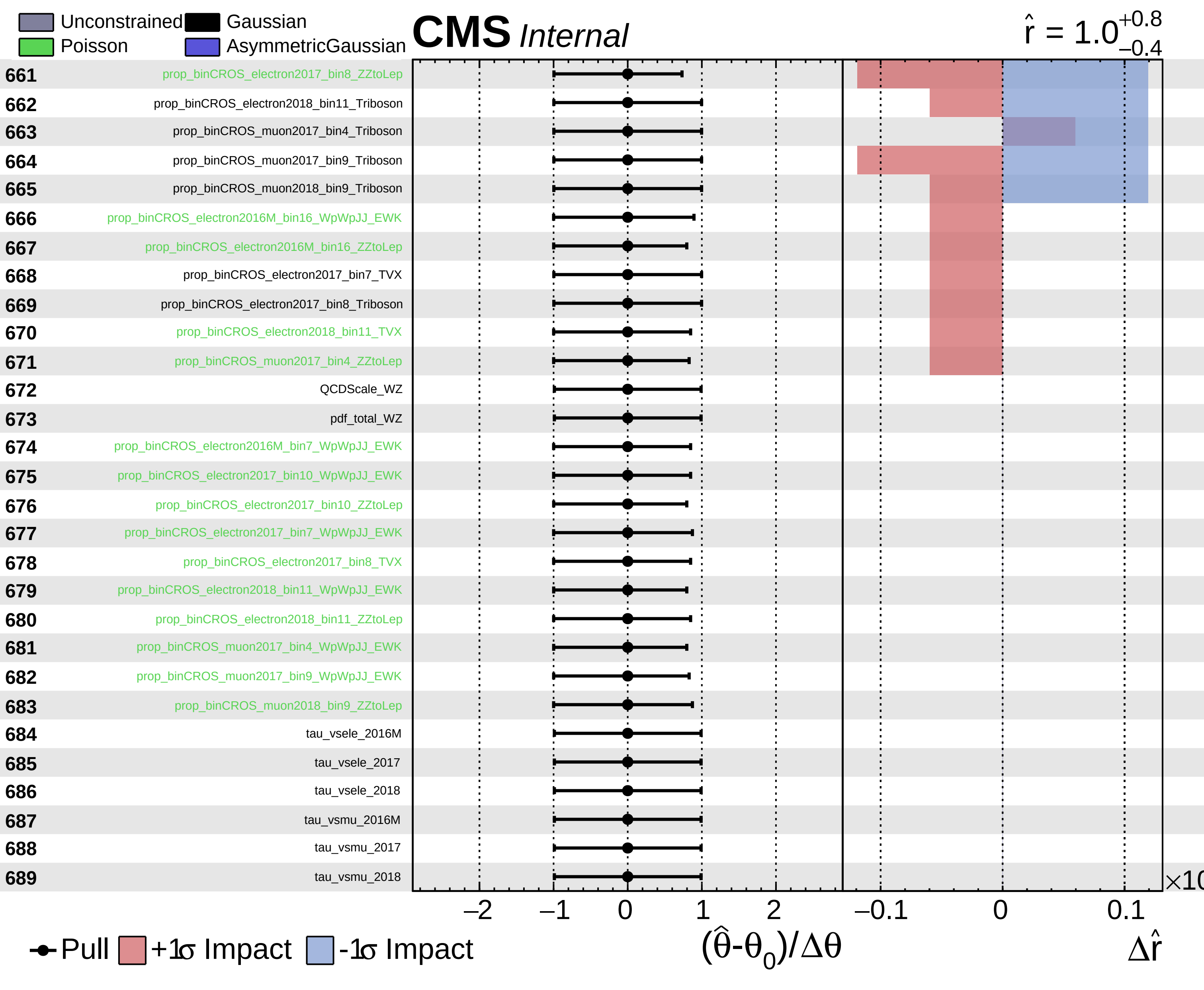


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
571 prop_binCRF_electron2018_bin16
572 prop_binCRF_electron2016M_bin8_WrongSign
573 prop_binSR_muon2017_bin17_WrongSign
574 prop_binCRF_muon2016M_bin15
575 prop_binCRTT_muon2016M_bin17
576 prop_binSR_electron2016M_bin2_VG
577 prop_binCRTT_electron2016M_bin17
578 prop_binCROS_electron2018_bin11_TTTo2L2Nu
579 prop_binCRF_electron2017_bin11_WpWpJJ_QCD
580 prop_binCROS_muon2017_bin13
581 prop_binCRF_electron2016M_bin8_TVX
582 prop_binCRF_electron2016M_bin9_ZZtoLep
583 prop_binCROS_electron2017_bin15
584 prop_binCRF_electron2017_bin12_WpWpJJ_QCD
585 prop_binCROS_muon2018_bin9_TTTo2L2Nu
586 prop_binCRF_muon2017_bin11_WpWpJJ_QCD
587 prop_binCROS_electron2018_bin14_TTTo2L2Nu
588 prop_binCRF_electron2017_bin16
589 prop_binCRF_muon2016M_bin11_WpWpJJ_QCD
590 prop_binCRF_muon2017_bin16
591 prop_binCRF_electron2016M_bin8_WpWpJJ_QCD
592 prop_binCROS_muon2016M_bin17
593 prop_binCRF_muon2017_bin11_Triboson
594 prop_binCRF_muon2016M_bin11_Triboson
595 prop_binCROS_electron2016M_bin13
596 prop_binSR_electron2016M_bin2_ZZtoLep
597 prop_binCROS_electron2016M_bin16_TTTo2L2Nu
598 prop_binCROS_electron2017_bin10_TTTo2L2Nu
599 prop_binCRF_electron2016M_bin9_TVX
600 prop_binCROS_electron2016M_bin15
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.8 −0.4
601 prop_binCRF_electron2017_bin12_Triboson
602 prop_binCRF_electron2016M_bin8_Triboson
603 prop_binCROS_electron2017_bin17
604 prop_binCROS_electron2017_bin7_TTTo2L2Nu
605 prop_binCRF_electron2016M_bin9_WpWpJJ_EWK
606 prop_binCROS_muon2017_bin4_TTTo2L2Nu
607 prop_binCROS_electron2018_bin11_VG
608 prop_binSR_electron2017_bin14_ZZtoLep
609 prop_binCROS_electron2016M_bin17_TTTo2L2Nu
610 prop_binCROS_electron2017_bin10_WZ
611 prop_binSR_electron2016M_bin17_ZZtoLep
612 prop_binCRF_muon2017_bin11_ZZtoLep
613 prop_binCROS_electron2017_bin7_WZ
614 prop_binCROS_electron2017_bin10_TVX
615 prop_binSR_electron2016M_bin14_ZZtoLep
616 prop_binSR_muon2016M_bin14_ZZtoLep
617 prop_binCROS_electron2017_bin8_TTTo2L2Nu
618 prop_binSR_electron2016M_bin2_TVX
619 prop_binCROS_electron2018_bin14_TVX
620 prop_binCRF_electron2016M_bin14_TVX
621 prop_binCROS_muon2018_bin9_TVX
622 prop_binSR_electron2016M_bin17_WrongSign
623 prop_binSR_electron2016M_bin2_WpWpJJ_EWK
624 prop_binCROS_electron2018_bin14_Fake
625 prop_binCRF_electron2016M_bin9_Triboson
626 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
627 prop_binSR_electron2018_bin16_WrongSign
628 prop_binCRF_electron2016M_bin8_ZZtoLep
629 prop_binSR_electron2016M_bin2_Triboson
630 prop_binCRF_electron2016M_bin9_WpWpJJ_QCD
−2 −1 0 1 2
−0.02 0 0.02
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
631 prop_binSR_electron2016M_bin16_WrongSign
632 prop_binCROS_electron2016M_bin9_Fake
633 prop_binCROS_electron2016M_bin9_TTTo2L2Nu
634 prop_binCROS_electron2018_bin14_Triboson
635 prop_binCRF_electron2017_bin11_Triboson
636 prop_binCROS_electron2018_bin14_ZZtoLep
637 prop_binCROS_electron2016M_bin17_Triboson
638 prop_binCROS_muon2017_bin9_TVX
639 prop_binCROS_electron2016M_bin16_Triboson
640 prop_binCROS_electron2016M_bin7_VG
641 prop_binCROS_electron2018_bin14_WpWpJJ_EWK
642 prop_binCROS_muon2017_bin4_VG
643 prop_binCROS_electron2016M_bin16_TVX
644 prop_binCROS_electron2016M_bin9_Triboson
645 prop_binCROS_electron2017_bin10_Triboson
646 prop_binCROS_electron2016M_bin7_TVX
647 prop_binCROS_electron2016M_bin9_TVX
648 prop_binCROS_muon2017_bin4_WZ
649 FES
650 prop_binCROS_electron2016M_bin17_ZZtoLep
651 prop_binCROS_electron2017_bin7_ZZtoLep
652 prop_binCROS_muon2017_bin4_TVX
653 prop_binCROS_muon2017_bin9_ZZtoLep
654 prop_binCROS_electron2016M_bin17_TVX
655 prop_binCROS_electron2016M_bin7_ZZtoLep
656 prop_binCROS_electron2016M_bin9_ZZtoLep
657 prop_binCROS_electron2016M_bin17_WpWpJJ_EWK
658 prop_binCROS_electron2016M_bin7_Triboson
659 prop_binCROS_electron2016M_bin9_WpWpJJ_EWK
660 prop_binCROS_electron2017_bin7_Triboson
−2 −1 0 1 2
−4 −2 0 2 4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.8}_{-0.4}$
661 prop_binCROS_electron2017_bin8_ZZtoLep
662 prop_binCROS_electron2018_bin11_Triboson
663 prop_binCROS_muon2017_bin4_Triboson
664 prop_binCROS_muon2017_bin9_Triboson
665 prop_binCROS_muon2018_bin9_Triboson
666 prop_binCROS_electron2016M_bin16_WpWpJJ_EWK
667 prop_binCROS_electron2016M_bin16_ZZtoLep
668 prop_binCROS_electron2017_bin7_TVX
669 prop_binCROS_electron2017_bin8_Triboson
670 prop_binCROS_electron2018_bin11_TVX
671 prop_binCROS_muon2017_bin4_ZZtoLep
672 QCDScale_WZ
673 pdf_total_WZ
674 prop_binCROS_electron2016M_bin7_WpWpJJ_EWK
675 prop_binCROS_electron2017_bin10_WpWpJJ_EWK
676 prop_binCROS_electron2017_bin10_ZZtoLep
677 prop_binCROS_electron2017_bin7_WpWpJJ_EWK
678 prop_binCROS_electron2017_bin8_TVX
679 prop_binCROS_electron2018_bin11_WpWpJJ_EWK
680 prop_binCROS_electron2018_bin11_ZZtoLep
681 prop_binCROS_muon2017_bin4_WpWpJJ_EWK
682 prop_binCROS_muon2017_bin9_WpWpJJ_EWK
683 prop_binCROS_muon2018_bin9_ZZtoLep
684 tau_vsele_2016M
685 tau_vsele_2017
686 tau_vsele_2018
687 tau_vsmu_2016M
688 tau_vsmu_2017
689 tau_vsmu_2018
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$